

National 5 Practical Electronics Past Papers

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November 2025

1 Intro

Much like the Higher and National 5 Physics Past Paper documents from Mr Davie and recently updated by Mr White, this document was created in order to make it easier to find past paper questions, both for teachers and students. I will add new past papers as they are issued. If you spot any mistakes, or want to suggest any improvements, send me an email at sputnikjoe77@gmail.com.

2 How to Use

The table on the next page contains links to questions sorted by topic and year. Clicking on a link will take you to that question. The marking instructions follow directly after each question. To return to the table click on Question Table at the top or bottom of any page. Trying to navigate the document without doing this is tedious.

Before starting any past paper questions I recommend that you have paper copies of the Relationships Sheet and Data Sheet to avoid wasting time.

Topic	2017 SQP	2018	2019	2024	2025
Circuit Symbols and Functions	1a	1a	1a	1a, 7b	1a
Resistor Values	1b	1b	1b	1b	1b
Cable & Wiring	2	2	2b		6
Series/Parallel Circuits	3	3	3, 5b	3	2, 3
Logic Gates	4	4	4	4	4
Logic Testing		5			7a
AC/DC	5	6	6	8	8
Electromagnetism			2a		
Switching Circuits	6b, 7b	7a			9b
Voltage Dividers	7a	7b			9a
Special Circuits (Bi-stable, 555, Comparator)			7	5a, 9	
Circuit Simulation/Testing	6a,c		8a	6, 7c	7b
Safety	9a			2	
Circuit Errors		8	5a, 8b	5b, 7a	5
Circuit Cost		9			
Block Diagrams	8	10	9	10	10
Circuit/Layout Diagrams	9b	11	10	11	11

Total marks — 60
Attempt ALL questions

1. (a) The table gives information about some circuit components. Some of the boxes have been left blank.

Complete the table for the missing entries.

3

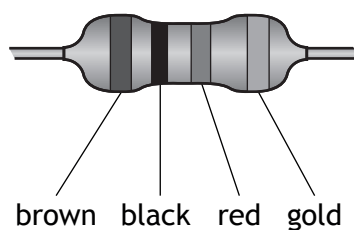
<i>Name</i>	<i>Symbol</i>	<i>Function</i>
resistor		limits current, drops voltage
		stores charge
bipolar transistor		electronic switch
diode		



* S 8 6 0 7 5 0 1 0 2 *

1. (continued)

(b) The diagram below shows the colour coding for a resistor.



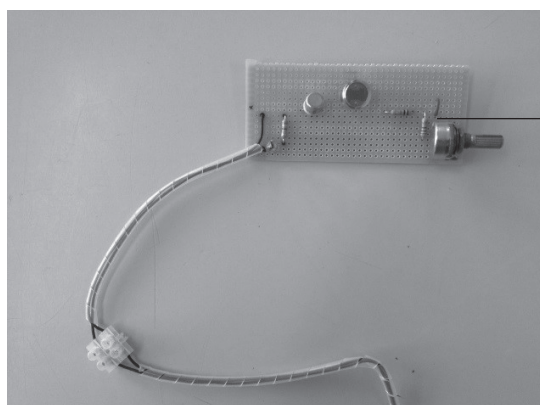
Using information in the data sheet:

- | | |
|---|---|
| (i) determine the resistance of the resistor; | 1 |
| (ii) state the tolerance in the resistance of the resistor; | 1 |
| (iii) determine the maximum and minimum resistance of the resistor. | 2 |

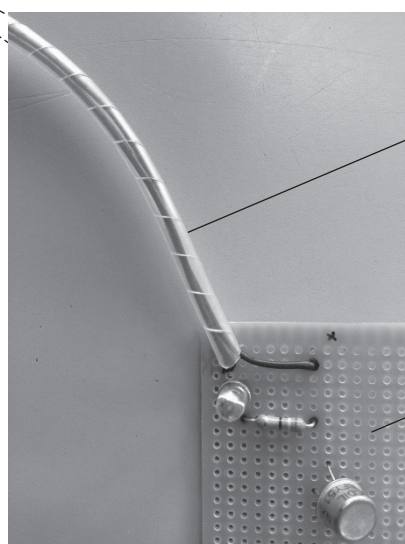
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2. The photographs show the wiring connections between a process board and an output board for a circuit.



process
board



wiring

output
board

- (a) Name the type of wiring shown which connects the two boards.

1

- (b) Describe how a continuity tester could be used to ensure good connections between the boards.

2



2. (continued)

- (c) State two methods that could be used to identify the correct wiring between the boards.

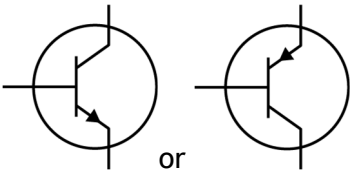
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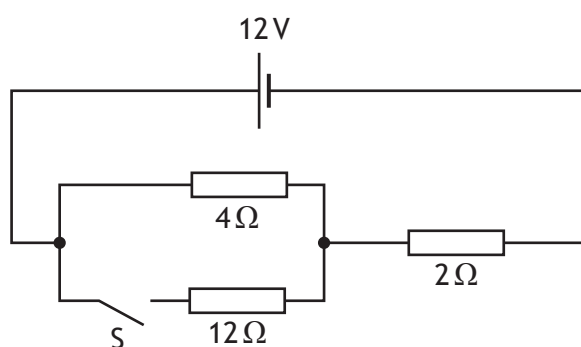


* S 8 6 0 7 5 0 1 0 5 *

Marking instructions for each question

Question			Answer	Max mark	Additional guidance
1.	(a)		capacitor 1  or 1 allows current to flow in only one direction or similar 1	3	Accept symbol for npn or pnp transistor (with or without circle).
	(b)	(i)	1000 Ω or 1 k Ω or 1k or 1k0	1	
		(ii)	5%	1	Percentage sign is required.
		(iii)	950 Ω or 950R 1 1050 Ω or 1k05 1	2	If unit omitted award a maximum of 1 mark.
2.	(a)		spiral wrap(ping)	1	
	(b)		connect to both ends of cable 1 lamp/LED lights 1	2	
	(c)		colour coding 1 numbering 1	2	

3. A student sets up a circuit as shown.



(a) Switch S is open.

(i) Calculate the total resistance of the circuit.

1

(ii) Calculate the current in the circuit.

3

(b) Switch S is now closed.

(i) Calculate the effective resistance of the two resistors in parallel.

3

(ii) Calculate the overall circuit resistance.

1



* S 8 6 0 7 5 0 1 0 6 *

4. Logic gates are widely used in electronic circuits.

(a) The truth table for a logic gate is shown.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Name the gate that produces this truth table.

1

(b) A logic gate is shown below.



Determine the logic state at X that would produce the output shown.

1

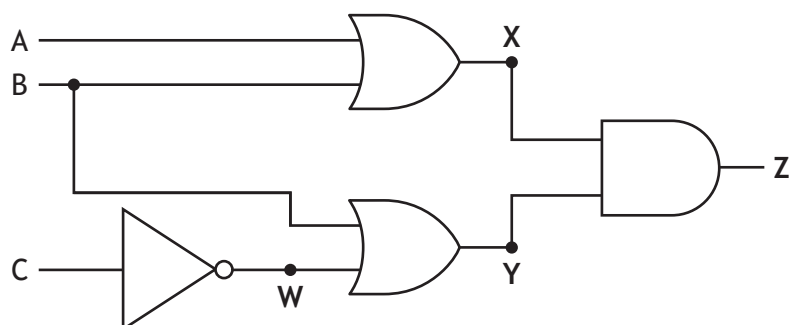
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* S 8 6 0 7 5 0 1 0 7 *

4. (continued)

(c)



Complete the truth table for the circuit.

4

A	B	C	W	X	Y	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(An additional truth table, if required, can be found on *page 18*.)

Question			Answer	Max mark	Additional guidance																																				
3.	(a)	(i)	6 Ω	1	Unit required.																																				
		(ii)	$V = IR$ 1 $12 = I \times 6$ 1 $I = 2 \text{ A}$ 1	3	Or consistent with (a)(i)																																				
	(b)	(i)	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ 1 $\frac{1}{R} = \frac{1}{4} + \frac{1}{12}$ 1 $R_T = 3 \text{ Ω}$ 1	3	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ is an acceptable alternative method. If wrong equation used, eg $R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$ then (0) marks Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{12} = 3 \text{ Ω}$ ↑ accept																																				
		(ii)	5 Ω	1	Unit required. Or consistent with (b)(i)																																				
4.	(a)		XOR or exclusive OR	1																																					
	(b)		1	1																																					
	(c)		<table><tr><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr></table> 1 mark for each correct column. Apply follow through between columns.	W	X	Y	Z	1	0	1	0	0	0	0	0	1	1	1	1	0	1	1	1	1	1	1	1	0	1	0	0	1	1	1	1	0	1	1	1	4	
W	X	Y	Z																																						
1	0	1	0																																						
0	0	0	0																																						
1	1	1	1																																						
0	1	1	1																																						
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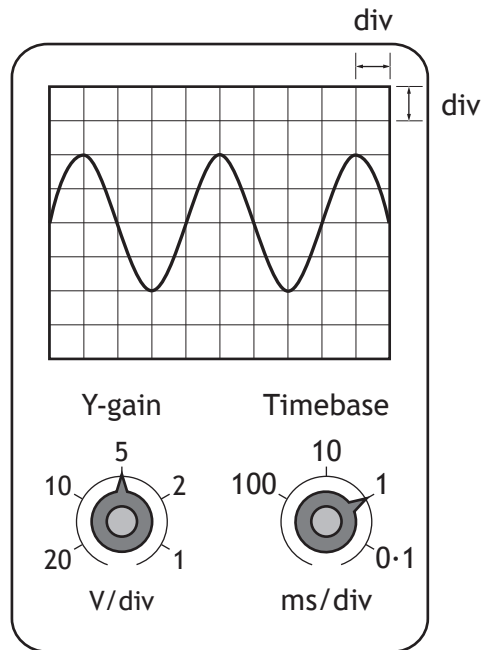
MARKS

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5. (a) The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown.



Determine the peak voltage of the signal.

1

[Turn over



* S 8 6 0 7 5 0 1 0 9 *

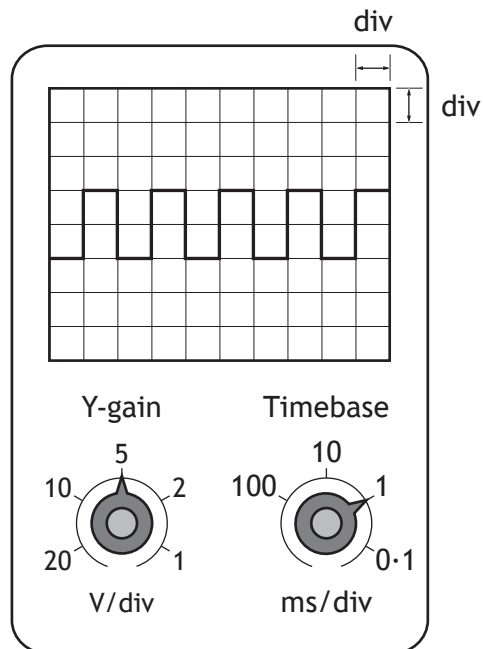
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5. (continued)

- (b) The output from the signal generator was changed to produce the trace shown below.

The Y-gain and timebase settings are also shown.



- (i) State whether the signal is analogue or digital.

1

- (ii) Determine the frequency of the signal.

3



MARKS

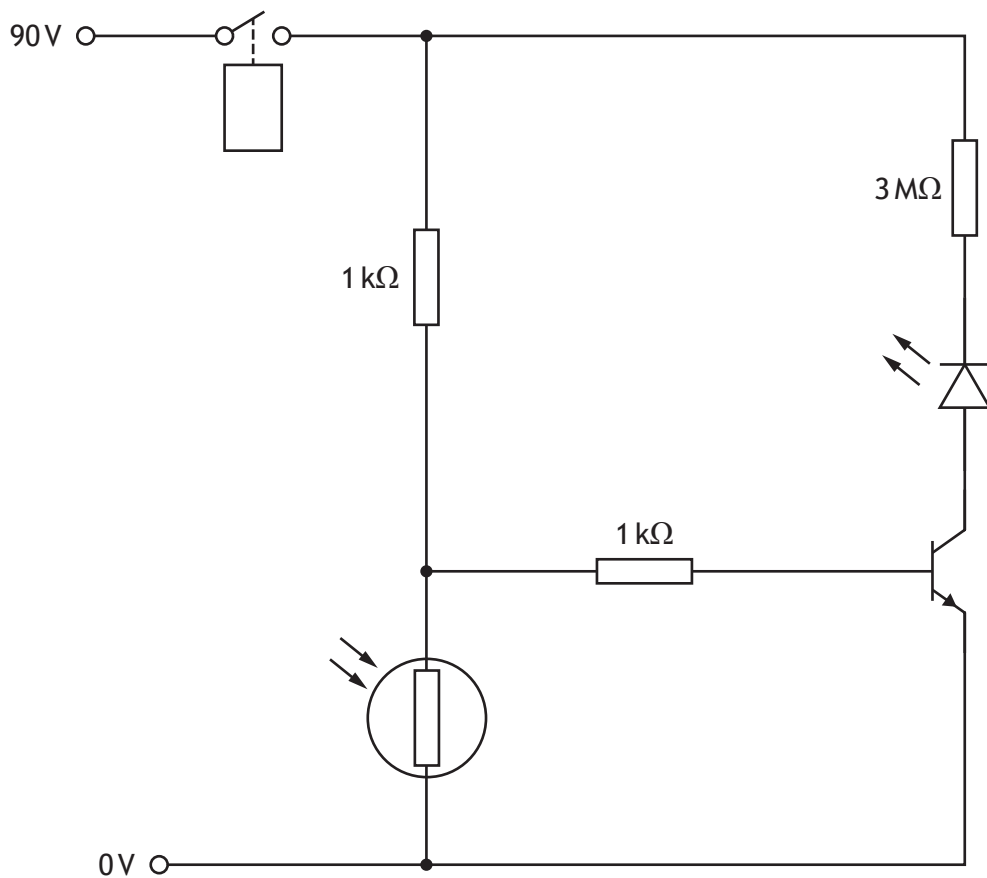
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6. Simulation is used extensively in circuit design.

(a) Give two reasons for simulating a circuit before it is constructed.

2

(b) A student simulated the circuit shown.



Identify the **three** errors in the circuit.

3

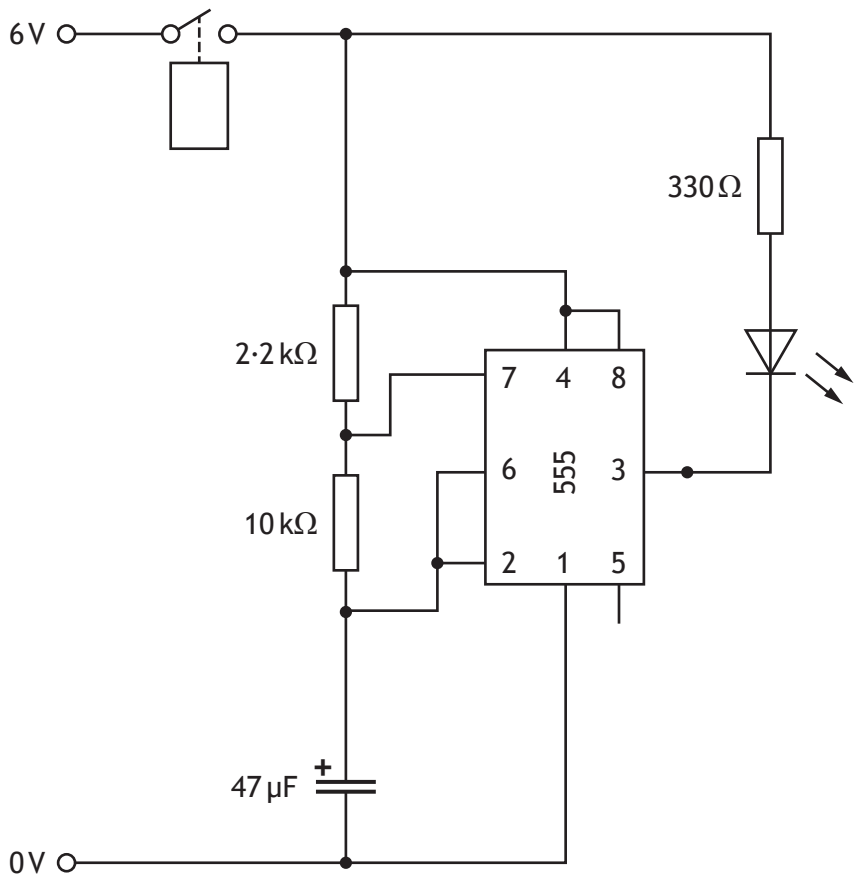
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* S 8 6 0 7 5 0 1 1 1 *

6. (continued)

(c) The student also simulated the 555 timer circuit shown below.



Complete the following table by giving **three** pre-power up checks for this circuit.

3

Pre-power up checklist
Check IC orientation

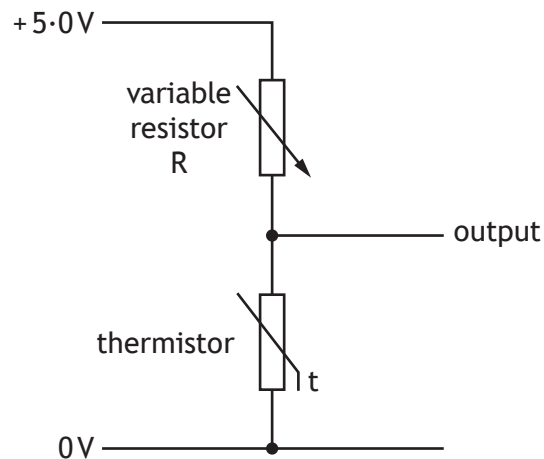


Question			Answer	Max mark	Additional guidance
5.	(a)		10 V	1	
	(b)	(i)	digital	1	
		(ii)	$T = (2 \times 1 \text{ ms}) = 2 \text{ ms}$ 1 $f = \frac{1}{T}$ 1 $f = 500 \text{ Hz}$ 1	3	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0.5 kHz, in which case 3 marks may be awarded.
6.	(a)		to establish if the circuit works to improve the circuit design easily edited to try different combinations allows testing of sub-systems Any other suitable response. Any two for 1 mark each.	2	Not: allows you to cost the circuit.
	(b)		incorrect/too large supply voltage 1 protective resistance for LED is too large 1 LED connected wrong way round 1	3	
	(c)		orientation of capacitor orientation of LED supply voltage resistance values Any three for 1 mark each.	3	

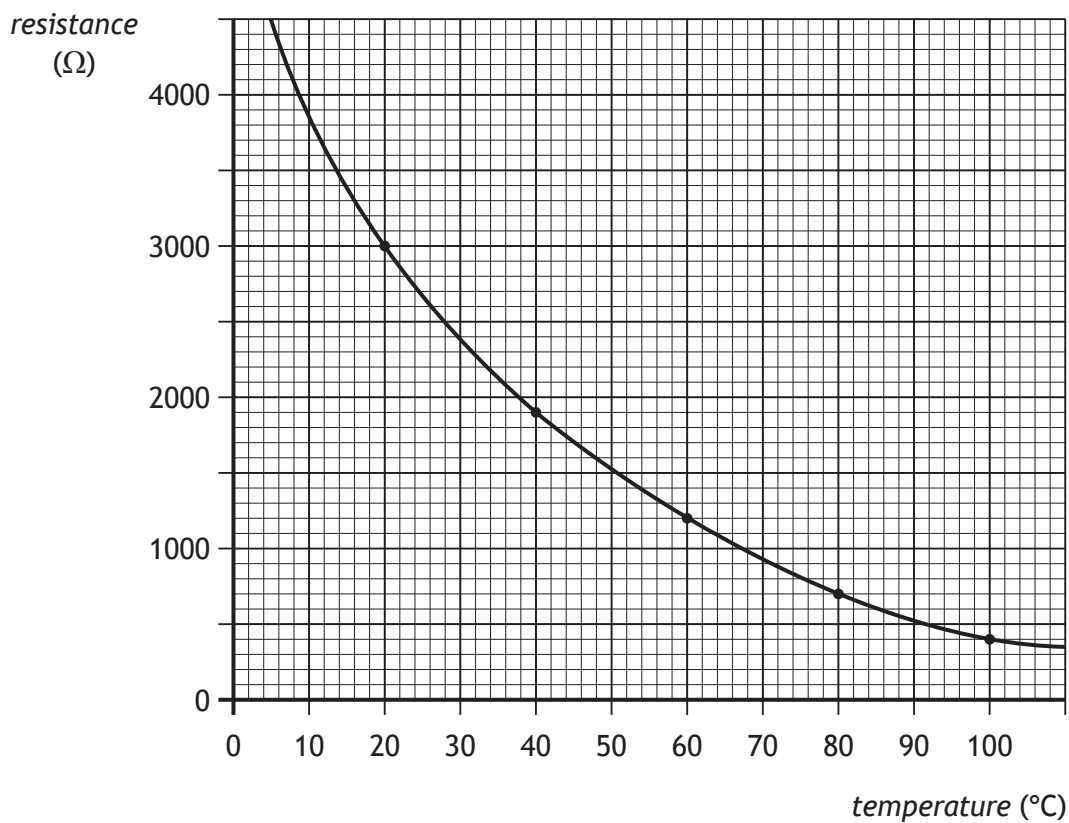
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7. A thermistor is used as a temperature sensor in a circuit to monitor and control the temperature of water in a tank. Part of the circuit is shown.



- (a) (i) The graph shows how the resistance of the thermistor varies with temperature.



Use the graph to determine the resistance of the thermistor when the temperature is 60 °C.

1

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* S 8 6 0 7 5 0 1 1 3 *

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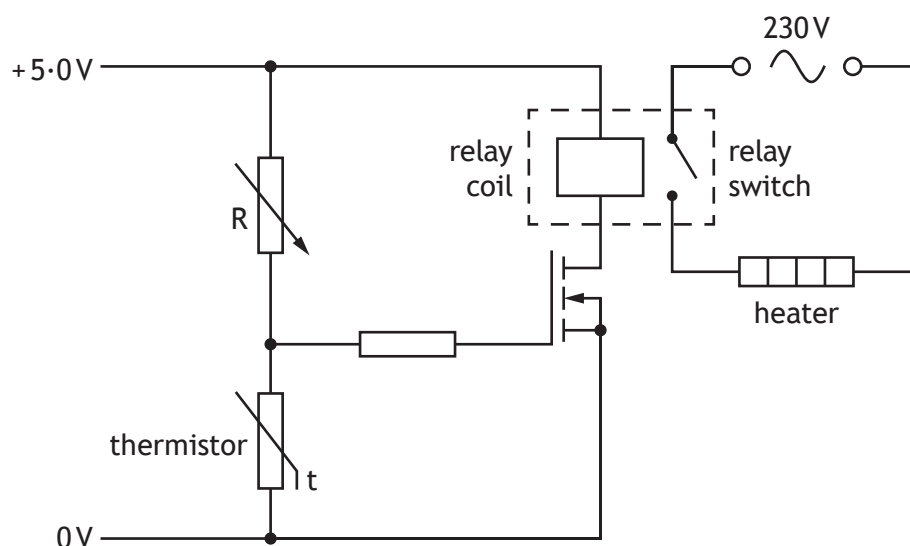
7. (a) (continued)

- (ii) The variable resistor R is now set at a resistance of $800\ \Omega$.

Calculate the voltage across the thermistor when the temperature is $60\ ^\circ\text{C}$.

3

- (b) The potential divider is now connected to a switching circuit to operate a heater.



- (i) Explain how the circuit operates to switch on the heater when the temperature falls below a certain value.

3

- (ii) The operation of this circuit can be improved by the addition of one component connected across the relay.

Name this component.

1



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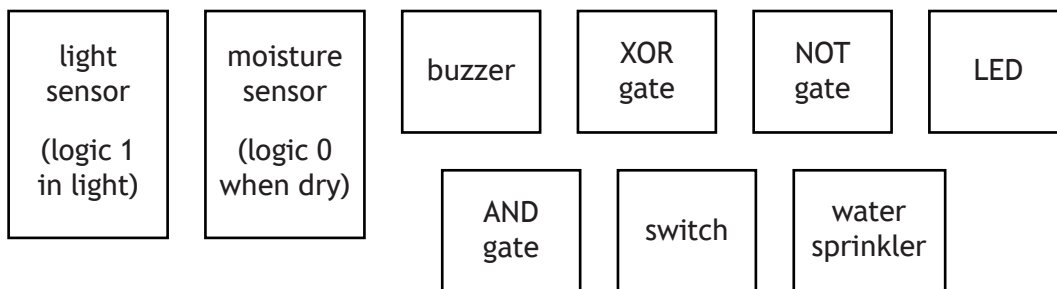
8. An engineer designs a system to control the moisture of the soil in a greenhouse.

The system must include a sensor which will measure the moisture content of the soil and turn on a water sprinkler system if the soil is too dry. An LED indicator should be included to show when the system is operating. There should also be a manual shut down switch to turn the system off when it is not required.

Selecting from the elements given below, draw a block diagram of an electronic solution that meets the engineer's criteria.

On your diagram, clearly indicate the input, process and output sections of your solution.

6



[Turn over



Question			Answer	Max mark	Additional guidance
7.	(a)	(i)	1200 Ω or 1k2	1	
		(ii)	$V_1 = \frac{R_1}{R_1 + R_2} \times V_s$ 1 $V_1 = \frac{1200}{1200 + 800} \times 5.0$ 1 $V_1 = 3 \text{ V}$ 1	3	Or consistent with (a)(i) Using Ohm's law: $R_T (= 1200 + 800) = 2000$ $I = \frac{V}{R}$ $I = \frac{5.0}{2000}$ $I = 2.5 \times 10^{-3} \text{ A}$ $V_1 = IR$ $V_1 = 2.5 \times 10^{-3} \times 1200$ $V_1 = 3 \text{ V}$ 1 mark for Ohm's Law relationship 1 mark for all substitutions 1 mark for final answer including unit
	(b)	(i)	Resistance of thermistor increases so voltage across it increases. 1 Transistor switches on at a certain voltage/2 V. 1 Relay activates and closes contacts in heater circuit. 1	3	If transistor switching voltage is given as 0.7 V award a maximum of 2 marks.
		(ii)	diode	1	
8.			1 mark for selecting both input devices 1 mark for selecting correct gates 1 mark for selecting both output devices 1 mark for correctly linking the inputs and gates 1 mark for correctly linking the outputs to the gates (arrows are NOT required on the connections) 1 mark for correctly identifying the three sections of the system	6	

MARKS

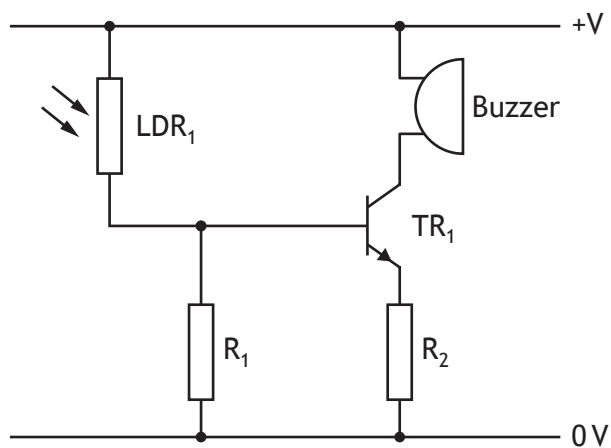
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9. (a) State two precautions that should be taken when soldering.

2

- (b) For the circuit diagram shown, complete the layout diagram.

5



Component data

R_1 – carbon film $10\text{ k}\Omega$ 0.25 W

R_2 – carbon film $1\text{ k}\Omega$ 0.25 W

LDR_1 – ORP12

TR_1 – BC108 npn bipolar

Buzzer – 6 V dc



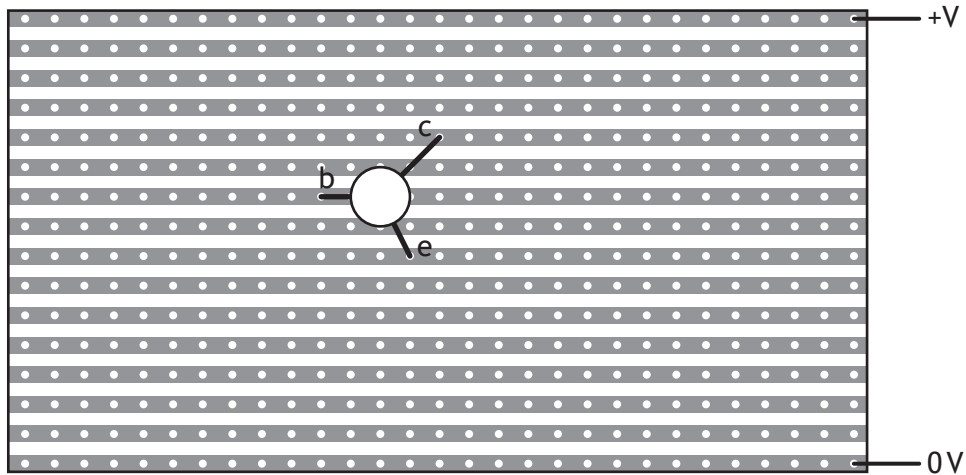
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9. (b) (continued)

Layout diagram



(Additional layout diagrams, if required, can be found on *page 19*.)

[END OF SPECIMEN QUESTION PAPER]



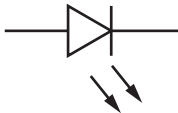
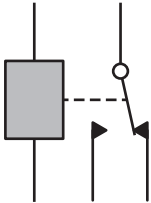
Question			Answer	Max mark	Additional guidance
9.	(a)		wear eye protection suitable ventilation avoid breathing in fumes Any other sensible suggestion. Any two for 1 mark each.	2	
	(b)		1 mark for each of the correctly positioned components. 1 mark for labelling all of the components.	5	‘footprint’ of components should be of an appropriate size

[END OF SPECIMEN MARKING INSTRUCTIONS]

Total marks — 60
Attempt ALL questions

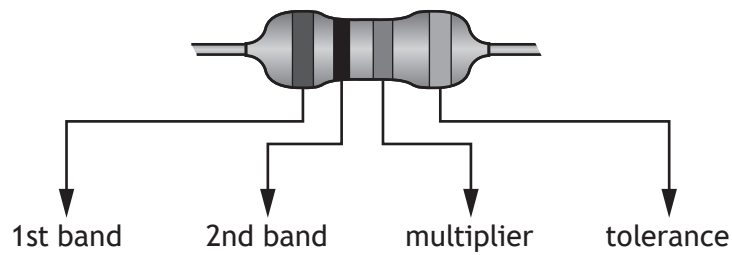
1. (a) The table gives information about some circuit components.
Some of the boxes have been left blank.
Complete the table for the missing entries.

3

Name	Symbol	Function
electrolytic capacitor		stores charge
		emits light
relay		
741 operational amplifier (op-amp)		comparator

1. (continued)

(b) The following diagram shows the colour coding for a resistor.



A student is comparing two resistors R_1 and R_2 . The colour code for each resistor is given in the table below.

<i>Resistor</i>	<i>1st band</i>	<i>2nd band</i>	<i>Multiplier</i>	<i>Tolerance</i>
R_1	brown	red	orange	silver
R_2	brown	red	orange	brown

Using information from the data sheet:

- (i) determine the resistance of resistor R_1 ; 1
- (ii) state the percentage tolerance value of resistor R_2 ; 1
- (iii) determine the colour codes for a resistor of value $6K8 \pm 5\%$. 2

<i>1st band</i>	<i>2nd band</i>	<i>Multiplier</i>	<i>Tolerance</i>
			gold

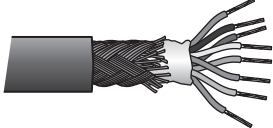
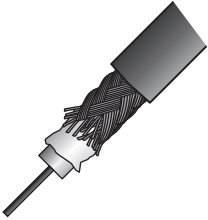
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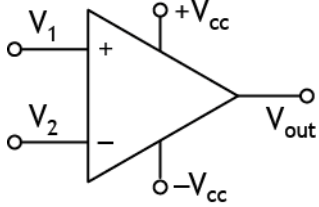
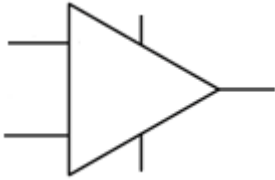
2

2. Complete the table by stating a typical use for each cable type shown.

	Cable type	Typical use
	multi-strand	
	coaxial	

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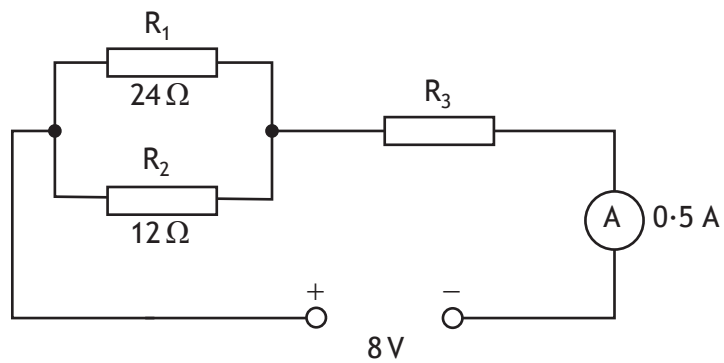
Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Light emitting diode/LED (1 mark) Electromagnetic switch (1 mark)  (1 mark)	3	Relay- explains control of high power/current/voltage circuit via a low power/current/voltage circuit. Minimum acceptable 741 diagram 
	(b)	(i)	12 000 Ω / 12 k Ω / 12K	1	
		(ii)	($\pm$) 1 (%)	1	
		(iii)	Blue and grey (1 mark) Red (1 mark)	2	1 mark for first and second bands correct. 1 mark for multiplier.
2.			Multi-strand: phone/telecom (1 mark) Coaxial: aerial/TV signal/radio signals/connecting CRO/audio (1 mark)	2	Accept any sensible suggestions.

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3. A student sets up the circuit shown.



- (a) Calculate the total resistance of the circuit.

3

Space for working and answer

- (b) Calculate the effective resistance of R_1 and R_2 in parallel.

3

Space for working and answer

- (c) Determine the resistance of resistor R_3 .

1

Space for working and answer



* X 8 6 0 7 5 0 1 0 6 *

3. (continued)

(d) Determine the voltage across R_3 .
Space for working and answer

1

(e) Calculate the power dissipated in resistor R_3 .
Space for working and answer

3

[Turn over

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Question			Expected response	Max mark	Additional guidance
3.	(a)		$V = IR$ (1 mark) $8 = 0.5 \times R$ (1 mark) $R = 16 \Omega$ (1 mark)	3	Accept 16R
	(b)		$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ (1 mark) $\frac{1}{R_T} = \frac{1}{24} + \frac{1}{12}$ (1 mark) $R_T = 8 \Omega$ (1 mark)	3	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ is an acceptable alternative method. If wrong equation used, eg $R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{24} = 8 \Omega$ $\uparrow$ accept Accept 8R
	(c)		$R_3 = 8 \Omega$	1	Or consistent with (a) and (b). Accept 8R
	(d)		$V_3 = 4 \text{ V}$	1	Or consistent with (a), (b), and (c).
	(e)		$P = \frac{V^2}{R}$ (1 mark) $P = \frac{4^2}{8}$ (1 mark) $P = 2 \text{ W}$ (1 mark)	3	Or consistent with (a), (b), (c), and (d). $P = IV$ 1 $P = 0.5 \times 4$ 1 $P = 2 \text{ W}$ 1 Or $P = I^2 R$ 1 $P = 0.5^2 \times 8$ 1 $P = 2 \text{ W}$ 1

4. Logic gates are widely used in electronic circuits.

(a) Complete the truth table for an OR gate.

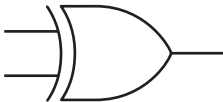
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A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 24*.)

(b) Name the logic gate shown below.

1



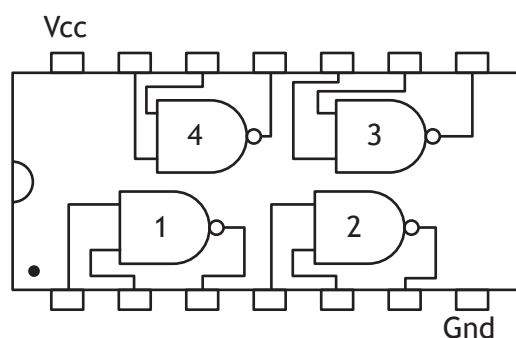
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4. (continued)

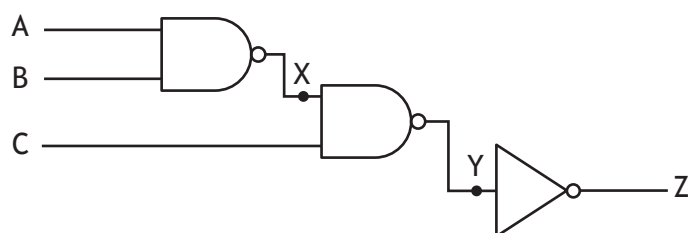
- (c) Using information from the data sheet, identify the **integrated circuit (IC)** shown.

1



- (d) Complete the truth table for the logic circuit shown.

3



A	B	C	X	Y	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 24*.)



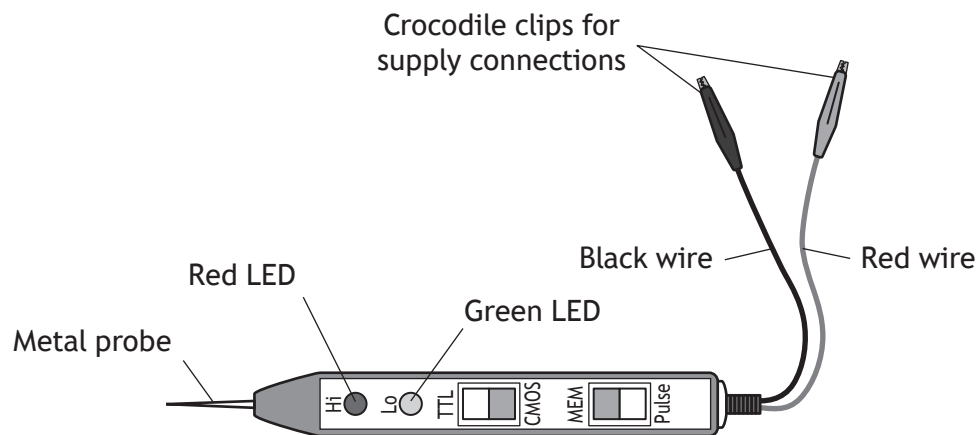
* X 8 6 0 7 5 0 1 0 9 *

Question			Expected response				Max mark	Additional guidance																																																						
4.	(a)		<table><tr><td>A</td><td>B</td><td>Output</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>				A	B	Output	0	0	0	0	1	1	1	0	1	1	1	1	1																																								
			A	B	Output																																																									
			0	0	0																																																									
			0	1	1																																																									
			1	0	1																																																									
1	1	1																																																												
	(b)		Exclusive OR / XOR				1																																																							
	(c)		7400 (quad 2 input NAND gates) OR Quad 2 input NAND (gates)				1	Do not accept NAND on its own.																																																						
	(d)		<table><tr><td>A</td><td>B</td><td>C</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>				A	B	C	X	Y	Z	0	0	0	1	1	0	0	0	1	1	0	1	0	1	0	1	1	0	0	1	1	1	0	1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	0	0	1	0	1	1	1	0	1	0	3	1 mark for each column. Apply follow through between columns.
A	B	C	X	Y	Z																																																									
0	0	0	1	1	0																																																									
0	0	1	1	0	1																																																									
0	1	0	1	1	0																																																									
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1	0	1	1	0	1																																																									
1	1	0	0	1	0																																																									
1	1	1	0	1	0																																																									
5.			Connect red wire of probe to V_{supply} and black wire to 0 V or equivalent. (1 mark)				3	Do not accept: connect the probe to the supply rails (on its own). 3 independent marks																																																						
			Place the probe tip onto circuit at selected point. (1 mark)																																																											
			Red LED lights. (1 mark)																																																											

MARKS

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5. A logic probe is used to test the inputs and outputs of a 74 series logic chip. The logic probe is set to TTL and pulse.



Describe how to connect and use the logic probe to test the inputs and outputs of the chip.

In your answer include how a logic 1 is detected.

3

[Turn over



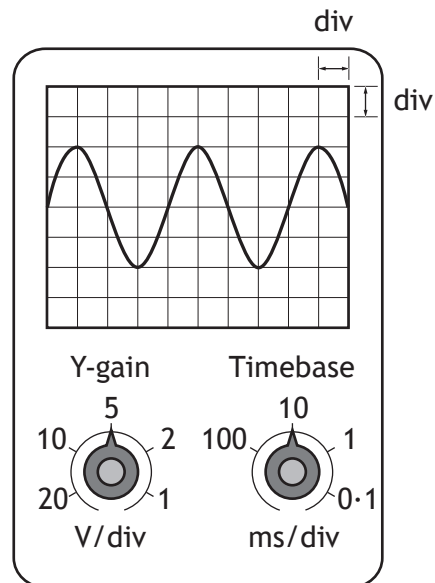
MARKS

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6. (a) The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown.



Determine the frequency of the signal.

3

Space for working and answer



MARKS

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6. (continued)

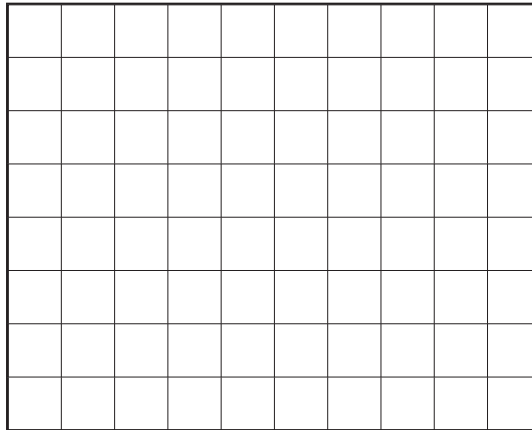
- (b) The peak voltage of the output signal from the signal generator is now **doubled**.

The frequency of the signal is unchanged.

The settings on the oscilloscope are unchanged.

Draw the new trace that would be shown on the screen.

1



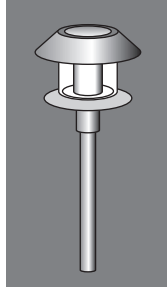
(An additional diagram, if required, can be found on *page 24*.)

[Turn over

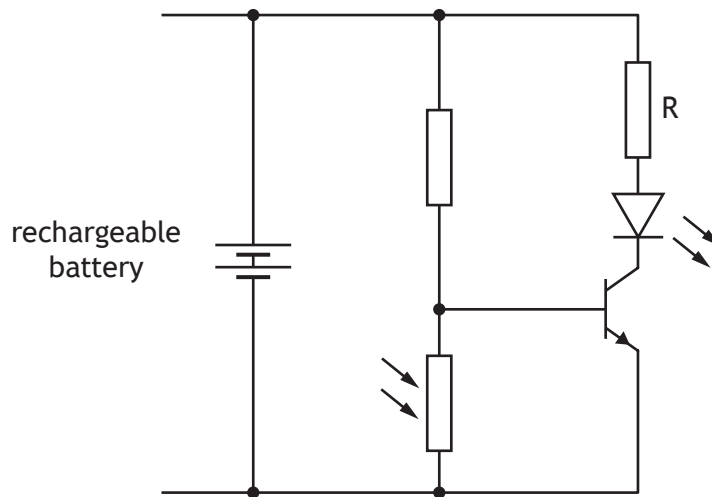


* X 8 6 0 7 5 0 1 1 3 *

7. A high intensity LED is used as a garden light. The light turns on automatically when it becomes dark.



(a) The LED is switched on using the following circuit.

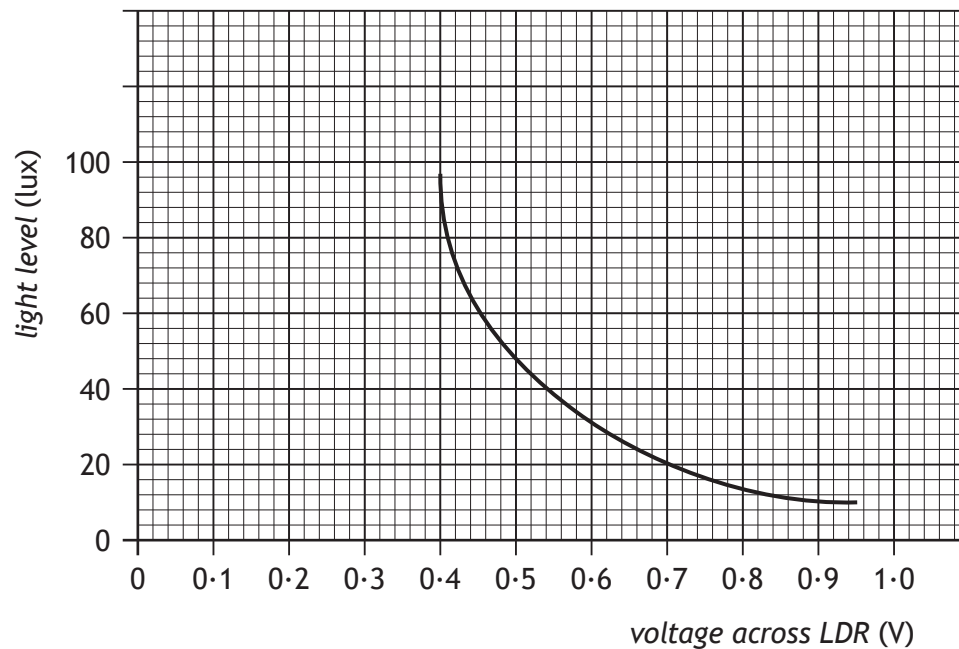


MARKS

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7. (a) (continued)

The graph shows the voltage across the LDR in this circuit for different light levels.



- (i) For the LED to switch on, the voltage across the LDR must be at least 0.7 V.

Determine the light level at which the LED switches on.

1

- (ii) Explain the purpose of resistor R.

1

- (iii) The manufacturer has used a bipolar transistor, as a MOSFET cannot be used in this circuit.

Explain why a MOSFET cannot be used in this circuit.

1

[Turn over

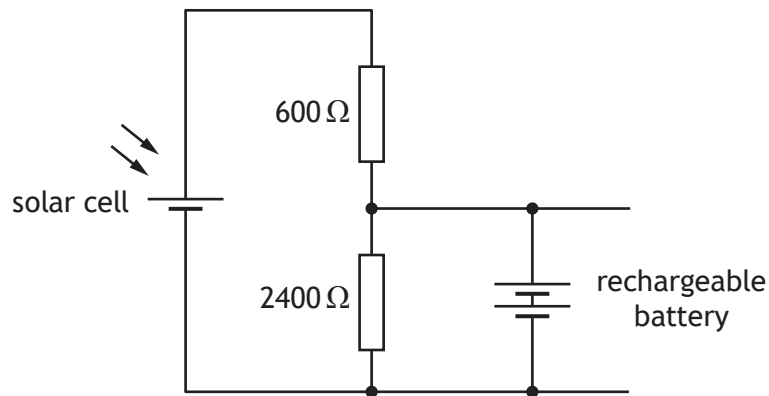


* X 8 6 0 7 5 0 1 1 5 *

7. (continued)

- (b) The light also contains a solar cell which charges the rechargeable battery during daylight hours.

Part of the circuit is shown.



At a particular light level, the voltage generated by the solar cell is 1.5 V .

Calculate the voltage across the rechargeable battery at this light level.

3

Space for working and answer



Question			Expected response	Max mark	Additional guidance
6.	(a)		$T = (4 \times 10 \text{ ms}) = 40 \text{ ms}$ (1 mark) $f = \frac{1}{T}$ (1 mark) $f = 25 \text{ Hz}$ (1 mark)	3	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0.025 kHz, in which case 3 marks may be awarded.
	(b)		Trace should have double the amplitude (with 3 peaks and two troughs).	1	
7.	(a)	(i)	20 lux	1	
		(ii)	Drop voltage/reduce current/protect LED/limit power.	1	Do not accept: voltage 'through' or current 'across' on their own.
		(iii)	Switch on voltage too high/2 V.	1	
	(b)		$V_2 = \frac{R_2}{R_1 + R_2} \times V_s$ (1 mark) $V_2 = \frac{2400}{600 + 2400} \times 1.5$ (1 mark) $V_2 = 1.2 \text{ V}$ (1 mark)	3	$V = IR$ $1.5 = I \times 3000$ $I = 5 \times 10^{-4} \text{ (A)}$ $V = IR$ $V = 5 \times 10^{-4} \times 2400$ $V = 1.2 \text{ V}$ 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. If correct answer given with no working award 3 marks.

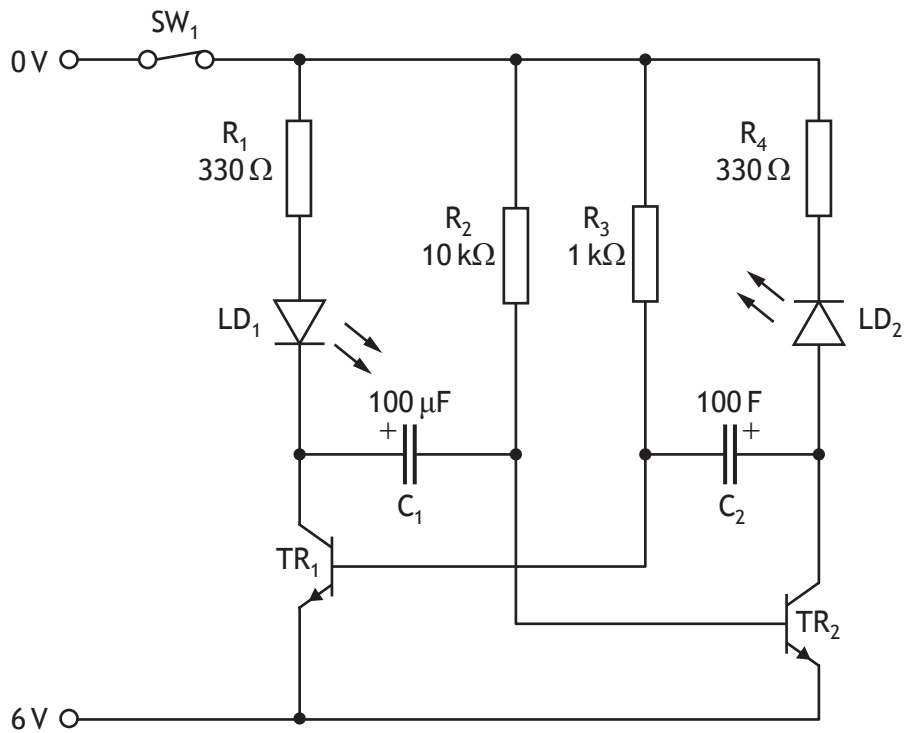
MARKS

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8. When switched on, the circuit shown should have the LEDs alternately flashing at the same rate. However the student has made four errors.

Identify the four errors.

4



Error 1:

Error 2:

Error 3:

Error 4:

[Turn over

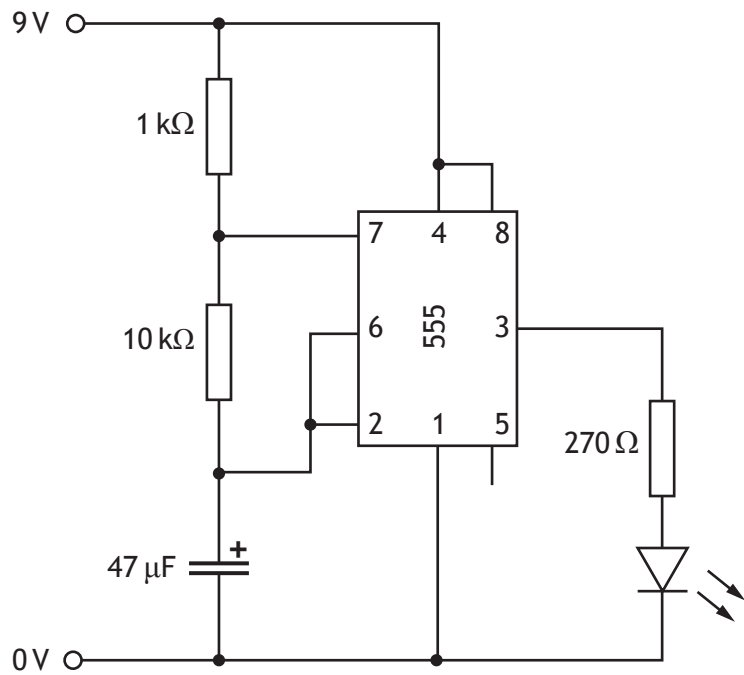


* X 8 6 0 7 5 0 1 1 7 *

MARKS

DO NOT WRITE IN THIS MARGIN

9. A student builds the circuit shown.



Using the information from two suppliers' catalogues shown on the opposite page, complete the costings sheet to produce the **lowest** cost for the circuit.

5

Supplier	Component	Product code	Cost (p)
JIMSON	NE555	TC124	20
SWIFT	8 way DIL socket	SK-0080	10
	LED 5 mm std		
	47 µF electrolytic capacitor		
	270R		
	1K		
	10K		



9. (continued)

Supplier	SWIFT			
	Component	Description	Product code	Cost
Integrated circuits	LM555CM	timers	IC-0283	45p
	NE555	timers	IC-0254	32p
	NE556	timers	IC-0216	25p
	8 way dil	ic socket	SK-0080	10p
Semi-conductors	LED	5 mm std red	SC-0155	4p
	LED	10 mm std red	SC-0177	10p
Electrolytic capacitors	10 μ F	16 V	CP-0555	10p
	47 μ F	16 V	CP-0566	18p
	47 μ F	6 V	CP-0588	10p
	47 μ F	5 V	CP-0599	8p
Resistors	220R	0.25 W carbon film 5%	EC-0159	0.5p
	270R	0.25 W carbon film 5%	EC-0161	0.5p
	1K	0.25 W carbon film 5%	EC-0175	0.5p
	10K	0.25 W carbon film 5%	EC-0182	1.5p
	100K	0.25 W carbon film 5%	EC-0198	3p
Supplier	JIMSON			
	Component	Description	Product code	Cost
Integrated circuits	LM555CM	timers	TC 123	90p
	NE555	timers	TC 124	20p
	NE556	timers	TC 125	80p
	8 way dil	ic socket	SK 099	50p
Semi-conductors	LED	5 mm std red	LD345	12p
	LED	10 mm std red	LD346	20p
Electrolytic capacitors	10 μ F	16 V	EC 798	10p
	47 μ F	16 V	EC 799	14p
	47 μ F	6 V	EC 800	10p
	47 μ F	5 V	EC 801	8p
Resistors	220R	0.25 W carbon film 5%	FR 921	0.25p
	270R	0.25 W carbon film 5%	FR 922	0.25p
	1K	0.25 W carbon film 5%	FR 923	0.25p
	10K	0.25 W carbon film 5%	FR 924	2p
	100K	0.25 W carbon film 5%	FR 925	5p



* X 8 6 0 7 5 0 1 1 9 *

Question			Expected response	Max mark	Additional guidance																																
8.			Power supply reversed/ wrong way round. (1 mark) LD₂ wrong polarity/wrong way round. (1 mark) C₂ value incorrect/too big/ different. (1 mark) R₂ / R₃ value incorrect/ different. (1 mark)	4	Errors can be listed in any order. Battery not accepted																																
9.			<table><tr><td>Supplier</td><td>Component</td><td>Product code</td><td>Cost (p)</td></tr><tr><td>JIMSON</td><td>NE555</td><td>TC124</td><td>20</td></tr><tr><td>SWIFT</td><td>8 way DIL socket</td><td>SK-0080</td><td>10</td></tr><tr><td>SWIFT</td><td>LED 5mm std</td><td>SC-0155</td><td>4</td></tr><tr><td>JIMSON</td><td>47µF electrolytic capacitor</td><td>EC799</td><td>14</td></tr><tr><td>JIMSON</td><td>270R</td><td>FR922</td><td>0.25</td></tr><tr><td>JIMSON</td><td>1K</td><td>FR923</td><td>0.25</td></tr><tr><td>SWIFT</td><td>10K</td><td>EC-0182</td><td>1.5</td></tr></table>	Supplier	Component	Product code	Cost (p)	JIMSON	NE555	TC124	20	SWIFT	8 way DIL socket	SK-0080	10	SWIFT	LED 5mm std	SC-0155	4	JIMSON	47µF electrolytic capacitor	EC799	14	JIMSON	270R	FR922	0.25	JIMSON	1K	FR923	0.25	SWIFT	10K	EC-0182	1.5	5	1 mark for each correct row.
Supplier	Component	Product code	Cost (p)																																		
JIMSON	NE555	TC124	20																																		
SWIFT	8 way DIL socket	SK-0080	10																																		
SWIFT	LED 5mm std	SC-0155	4																																		
JIMSON	47µF electrolytic capacitor	EC799	14																																		
JIMSON	270R	FR922	0.25																																		
JIMSON	1K	FR923	0.25																																		
SWIFT	10K	EC-0182	1.5																																		

10. An engineer designs a system to control the temperature within a greenhouse.

The system includes a sensor which measures the temperature within the greenhouse.

If the temperature exceeds 24 °C this turns on a motor to open a window.

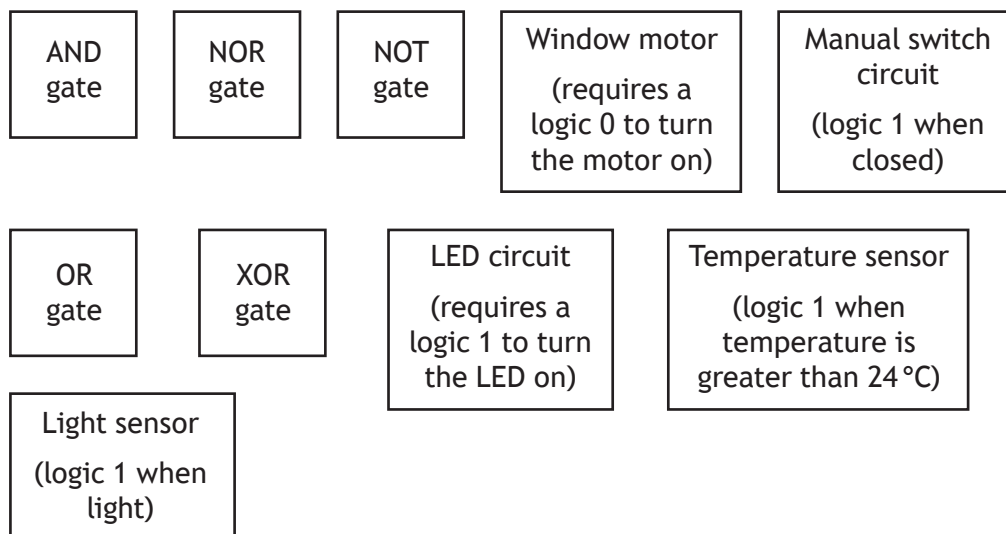
An LED indicator lights when the window is open.

There is also a manual switch to turn the whole system on and off.

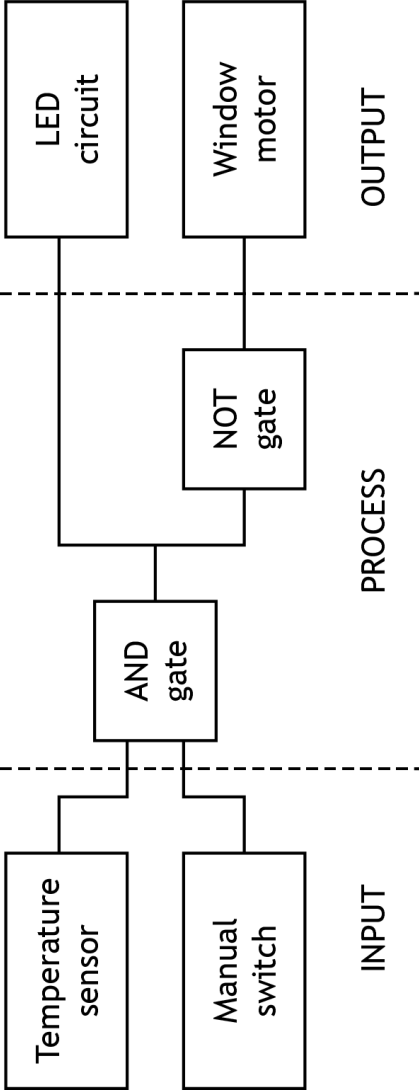
Selecting from the elements given, draw a block diagram of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution.

6



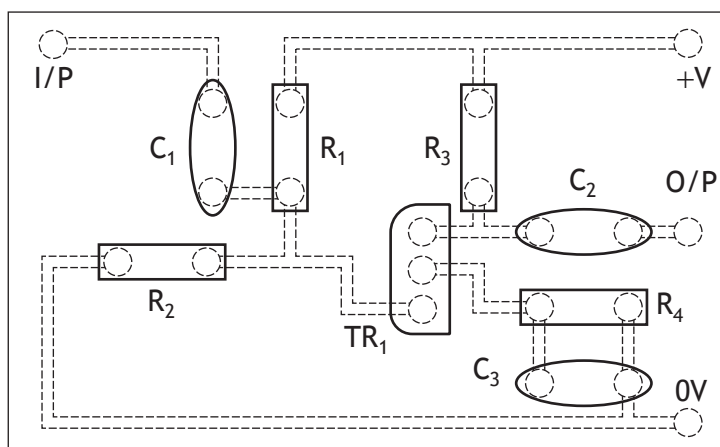
* X 8 6 0 7 5 0 1 2 0 *

Question	Expected response	Max mark	Additional guidance
10.	1 mark for selecting both input devices. 1 mark for selecting correct gates. 1 mark for selecting both output devices. 1 mark for correctly linking the inputs to gates. 1 mark for correctly linking the outputs to the gates. 1 mark for correctly identifying the three sections of the system diagram. 	6	All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections).

MARKS

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11. The following PCB layout shows a circuit with the following component data.



Component data

R_1 — carbon film 82K 0.25 W

R_2 — carbon film 10K 0.25 W

R_3 — carbon film 8K2 0.25 W

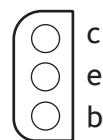
R_4 — carbon film 1K 0.25 W

C_1 — 0.1 μ F

C_2 — 0.1 μ F

C_3 — 10 μ F

TR_1 — BC182 npn bipolar



TR_1 pin connections

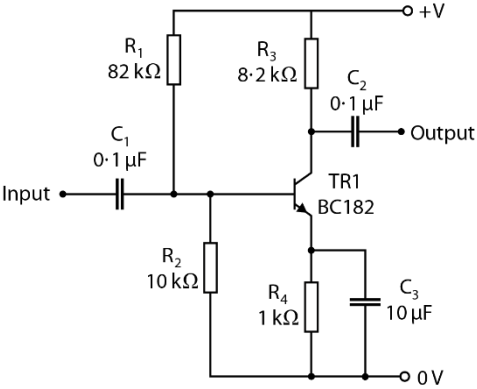
Draw a circuit diagram for this circuit.

Each component must be labelled.

6



* X 8 6 0 7 5 0 1 2 2 *

Question		Expected response	Max mark	Additional guidance
11.		1 mark for both power rails. 1 mark for transistor oriented in a functional manner and uses correct symbol. 1 mark for the four resistors (with correct symbol) relative to power rails and transistor. 1 mark for C3 in parallel to R4 and uses acceptable symbol. 1 mark for placement of input and C1(with correct symbol) relative to transistor. 1 mark for placement of C2 (with correct symbol) and output relative to transistor. 	6	Component symbols must be used. Accept electrolytic capacitor symbols. Each component must be identifiable. Accept a mixture of component values and identifiers. (accept R notation) Node spot for input/outputs are not required. Power rails must have V+ or Vs and 0V with  end node.

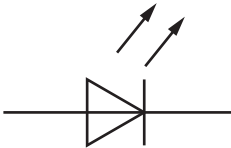
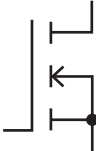
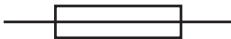
[END OF MARKING INSTRUCTIONS]

Total marks — 60

Attempt ALL questions

1. The table below gives information about some circuit components.
Some of the boxes have been left blank.
- (a) Complete the table for the missing entries.

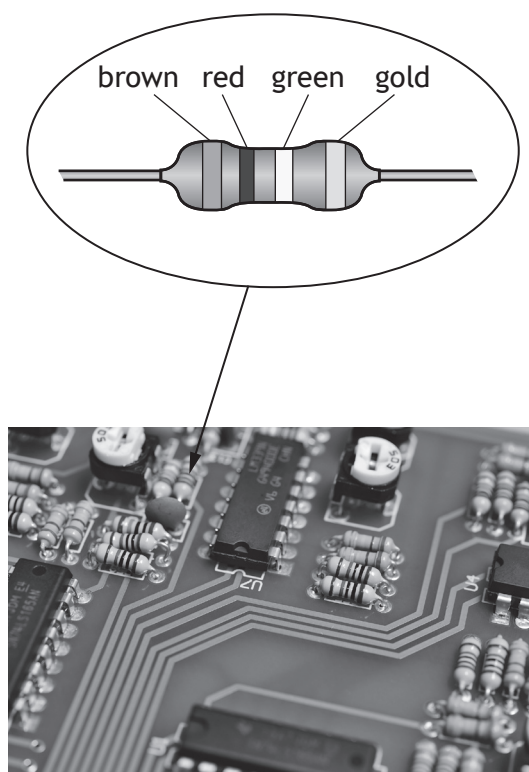
3

Component name	Symbol	Function
Light emitting diode		Emits light indicating a current flow
MOSFET transistor		
Motor		Converts electrical energy to kinetic energy
		To protect wiring from too much current

1. (continued)

- (b) A technician is examining the circuit board shown below when a fault is discovered.

A resistor with an incorrect value has been used.



Use the information in the data sheet to answer the questions below.

- (i) Determine the resistance of the resistor.

1

- (ii) The technician replaces the incorrect component with a resistor of value 600Ω which has a 2% tolerance.

Determine the minimum and maximum resistance of the resistor.

2



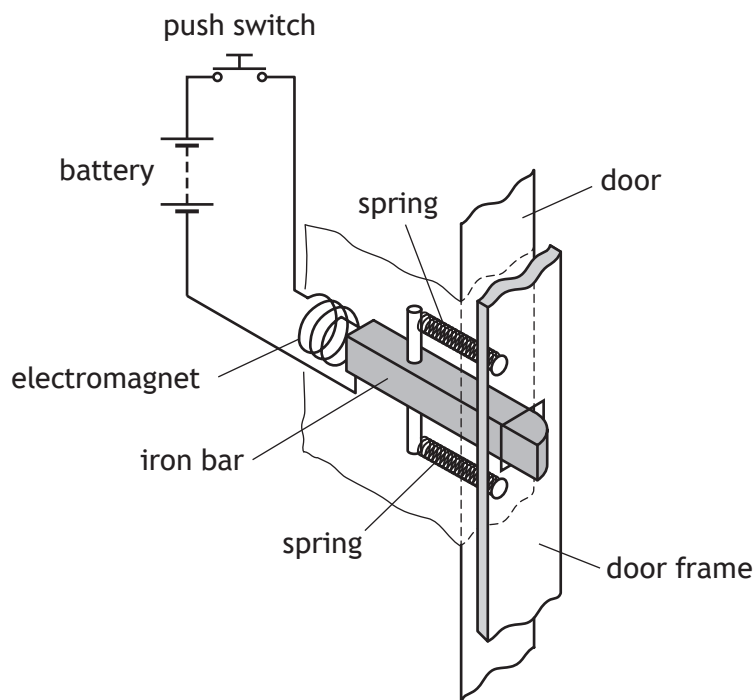
* X 8 6 0 7 5 0 1 0 3 *

MARKS

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2. A remote entry system for a block of flats allows a resident to unlock the outside door from inside their flat.

Part of this system is shown in the diagram below.



- (a) Explain how this part of the system operates to unlock the door.

2

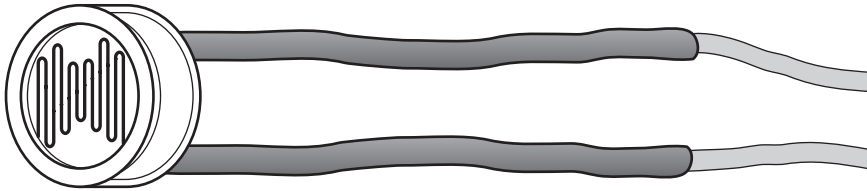


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2. (continued)

- (b) A light dependent resistor and sleeving is connected to a process board by two wires as shown in the picture below.



- (i) Name the wiring technique shown.

1

- (ii) State why this technique is used.

1

[Turn over



* X 8 6 0 7 5 0 1 0 5 *

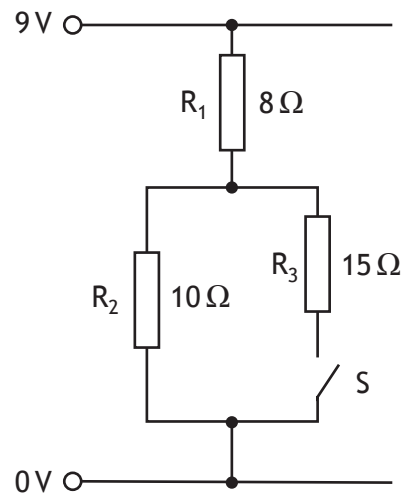
Marking instruction for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Switch electrical (1 mark)  (1 mark) FUSE (1 mark)	3	
	(b)	(i)	1M2 OR 1.2MΩ OR 1,200,000Ω	1	Also accept scientific notation with unit
		(ii)	Minimum 588Ω OR 588R (1 mark) Maximum 612Ω OR 612R (1 mark)	2	
2.	(a)		(When switch closed) current flows through electromagnet/coil (1 mark) The iron bar is attracted by magnetism to the electromagnet/coil (1 mark)	2	
	(b)	(i)	Heat shrink (tubing)	1	Not sleeving Accept flying leads
		(ii)	To avoid short circuiting the LDR/ stop the bare wires touching	1	If flying leads then for remote connection to other parts of circuitry

MARKS

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3. A circuit diagram is shown below.



Switch S is **open**.

- (a) (i) Calculate the total resistance of the circuit.

1

Space for working and answer

- (ii) Calculate the current in the circuit.

3

Space for working and answer



* X 8 6 0 7 5 0 1 0 6 *

MARKS

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3. (a) (continued)

(iii) Calculate the power dissipated in resistor R_1 .

3

Space for working and answer(b) Switch S is now **closed**.Calculate the effective resistance of R_2 and R_3 in parallel.

3

Space for working and answer

[Turn over]



* X 8 6 0 7 5 0 1 0 7 *

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	18Ω	1	Accept 18R
		(ii)	$V = IR$ (1 mark) $9 = I \times 18$ (1 mark) $I = 0.5 \text{ A}$ (1 mark)	3	Accept answer consistent with 3(a)(i) Answer must have unit
		(iii)	$P = I^2 R$ (1 mark) $P = 0.5^2 \times 8$ (1 mark) $P = 2 \text{ W}$ (1 mark)	3	Accept answer consistent with 3(a)(ii) Answer must have unit Accept $P=IV = v^2/R$ if full working given
	(b)		$\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3}$ (1 mark) $\frac{1}{R_T} = \frac{1}{10} + \frac{1}{15}$ (1 mark) $R_T = 6\Omega$ (1 mark)	3	$R_T = \frac{R_2 R_3}{R_2 + R_3}$ Is an acceptable alternative method If wrong equation used, eg $R_T = \frac{1}{R_2} + \frac{1}{R_3}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{15} + \frac{1}{10} = 6\Omega$ $\uparrow$ accept Accept = 6 R Accept any subscript on component resistors

4. There are many different types of logic gates used in electronic circuits.

(a) Draw the circuit symbol for a NAND gate.

1

(b) The truth table for a logic gate is shown below.

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

Name the logic gate that produces this truth table.

1

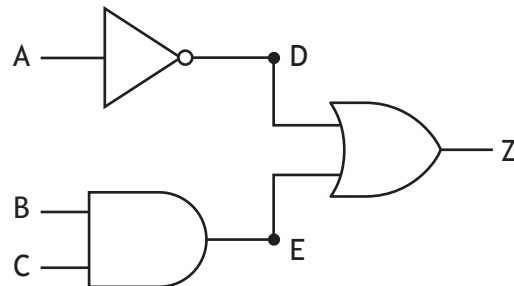


4. (continued)

Many electronic devices use several logic gates connected to one another.

(c) Complete the truth table for the logic circuit shown below.

3



A	B	C	D	E	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 20*)

[Turn over



* X 8 6 0 7 5 0 1 0 9 *

Question			Expected response	Max mark	Additional guidance																																																						
4.	(a)			1																																																							
	(b)		NOR (gate)	1																																																							
	(c)		<table border="1"> <thead> <tr> <th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>Z</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr> </tbody> </table>	A	B	C	D	E	Z	0	0	0	1	0	1	0	0	1	1	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1	0	0	0	0	0	1	0	1	0	0	0	1	1	0	0	0	0	1	1	1	0	1	1	3	1 mark per column Apply follow through between columns
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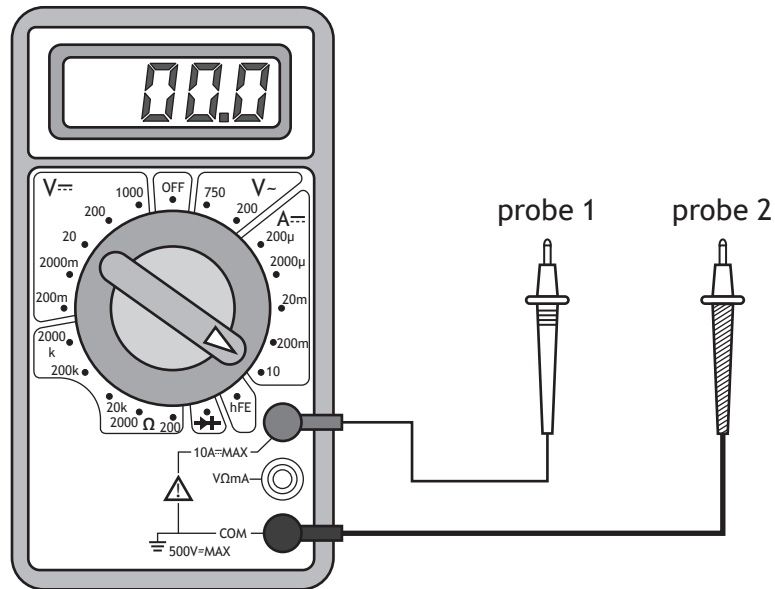
MARKS

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5. A student wants to measure the **voltage** across a component in a circuit and sets up the multimeter as shown below.

(a) Identify the two errors the student has made.

2



Error 1:

Error 2:



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5. (continued)

- (b) The student is given a resistor with no colour markings.

Describe how the student would set up and use a multimeter to accurately measure the resistance of the resistor.

3

[Turn over

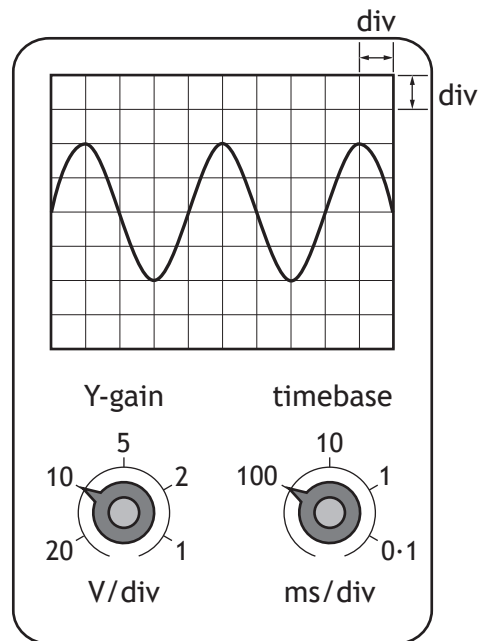


* X 8 6 0 7 5 0 1 1 1 *

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6. An oscilloscope is connected to show the output trace from a signal generator. The trace is shown on the screen. The Y-gain and timebase settings are also shown.



- (a) Determine the frequency of the signal.

3

Space for working and answer



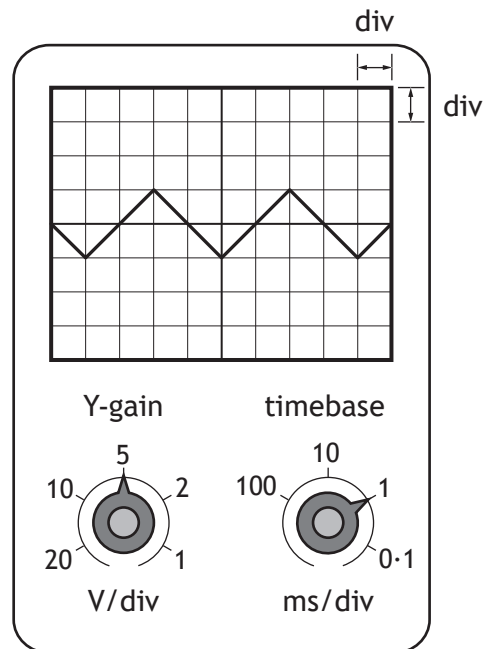
* X 8 6 0 7 5 0 1 1 2 *

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6. (continued)

The output from the signal generator was changed to produce the trace shown below.



(b) State whether the signal is analogue or digital.

1

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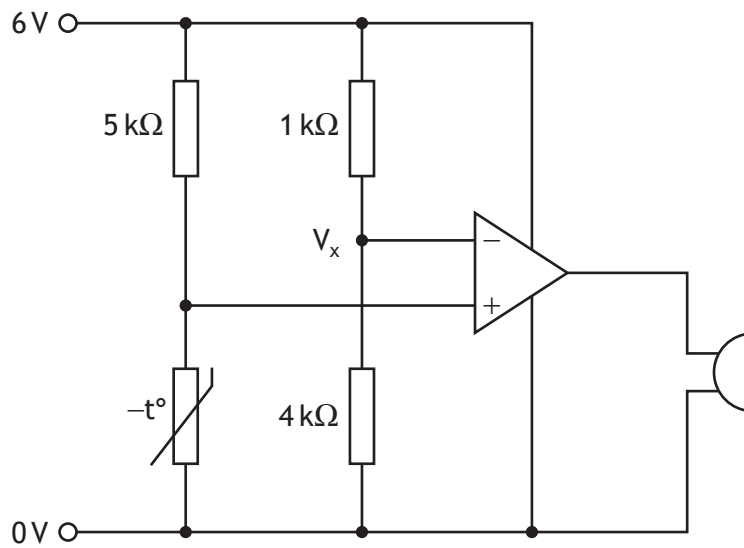
* X 8 6 0 7 5 0 1 1 3 *

Question			Expected response	Max mark	Additional guidance
5.	(a)		Error: meter on incorrect scale/ range/setting (1 mark) Error: probe 1 in wrong socket (1 mark) OR probe 1 should be in VΩmA socket OR middle/centre socket	2	Accept Meter on ampere/current scale Multi-meter should be set to $V_{\overline{\sim}}$ or $V_{\sim}$ Probe must be identified as probe 1
	(b)		Connect probe 1 to middle socket/ VΩmA socket. (Probe 2 to the COM socket) (1 mark) Set scale to Ω/ohms/resistance (1 mark) Start at highest resistance range and work down until display reads 1 then go up one range. (1 mark)	3	Accept range/scale as interchangeable Accept: probe 1 => COM Probe 2 => VΩmA Or until measured resistance is bigger than the next smallest scale/range
6.	(a)		Period $t = 400\text{ms}$ or 0.4s (1 mark) $f = \frac{1}{T}$ (1 mark) $f = \frac{1}{0.4}$ $f = 2.5\text{Hz}$ (1 mark)	3	Must have unit If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0.0025 kHz, in which case 3 marks may be awarded.
	(b)		Analogue (signal)	1	

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7. A low temperature warning system is controlled using a LM741 comparator circuit as shown in the simulation below.



- (a) Calculate the reference voltage V_x .

3

Space for working and answer

- (b) State the resistance of the thermistor when the voltage across it is equal to the reference voltage V_x .

1

- (c) Describe how this circuit works.

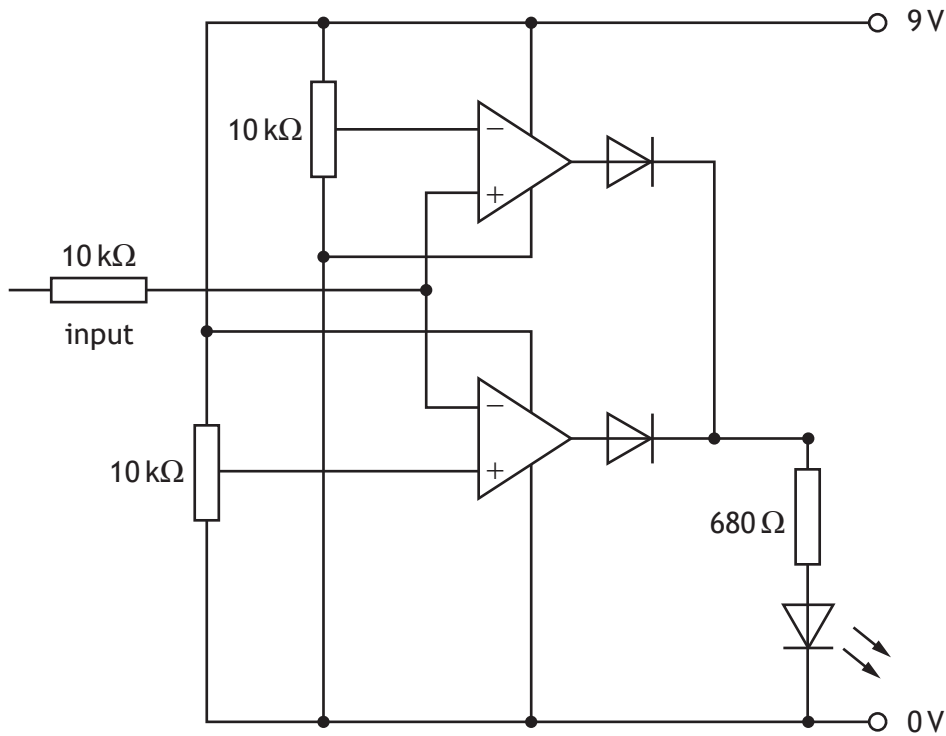
3



* X 8 6 0 7 5 0 1 1 4 *

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8. A student simulated the circuit shown below.



(a) Complete the following table by giving **three** pre-power up checks for this circuit.

3

Pre-power up checklist
Check both IC's orientation

[Turn over



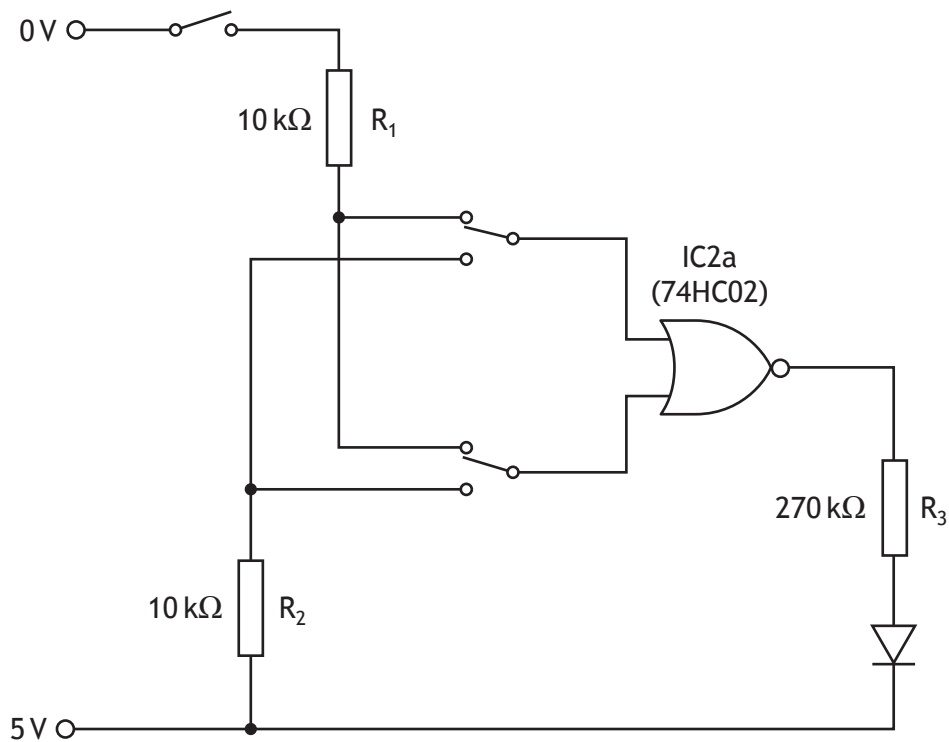
8. (continued)

- (b) A student also produced a simulation of a circuit to logic test a single AND gate using a LED as the output device.

However the simulation does not work as specified.

Identify **the four** errors in the simulation below.

4



Error 1:

Error 2:

Error 3:

Error 4:



* X 8 6 0 7 5 0 1 1 6 *

Question			Expected response	Max mark	Additional guidance
7.	(a)		$V_x = \frac{R_2}{R_1 + R_2} \times V_s \quad (1 \text{ mark})$ $V_x = \frac{4000}{5000} \times 6 \quad (1 \text{ mark})$ $V_x = 4.8V \quad (1 \text{ mark})$	3	Accept (4/5) x 6. 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. If correct answer given with no working award 3 marks.
	(b)		20K or 20kΩ or 20,000Ω	1	Must have unit or be in R notation
	(c)		As temperature decreases thermistor resistance increases. (1 mark) As thermistor resistance increases the voltage across the thermistor increase. (1 mark) When voltage across thermistor = Vx then LM741/comparator/IC/op amp switches on (1 mark)	3	Accept V ref = Vx = trigger voltage = switch on voltage = threshold (voltage) Accept buzzer/alarm sounds in place of LM741 switching on but not bell
8.	(a)		Any three from • check diode orientation • check LED orientation • correct value of resistors • both ICs are the same type • both potentiometers are of the same type (linear or log) • both potentiometers have same maximum resistance • check power supply connected correctly/right way round	3	Do not accept variable resistors.
	(b)		Voltage supply rails are reversed/wrong way around (1 mark) Incorrect logic gate used (1 mark) R ₃ resistance too large (1 mark) Using diode not LED (1 mark)	4	Accept using a NOR gate Accept not using a LED Not resistor too big/large If power not identified then accept diode/led wrong way round

MARKS

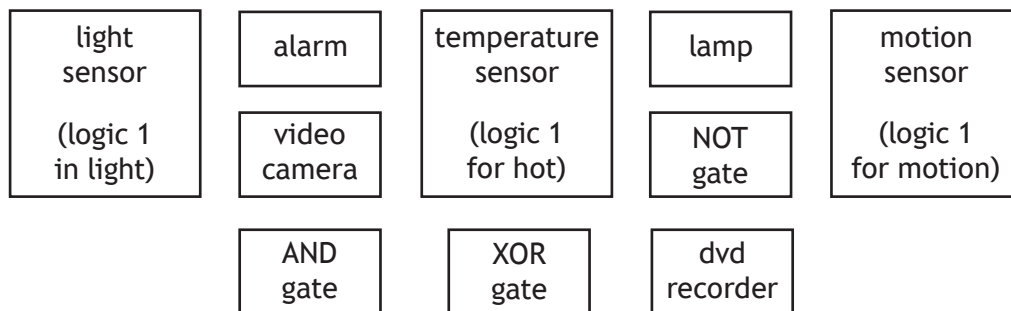
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9. A security system for a house requires a lamp and an alarm to turn on when movement is detected outside the house. The security system only needs to operate at night.

Selecting from the elements given below, draw a block diagram of an electronic solution for the security system.

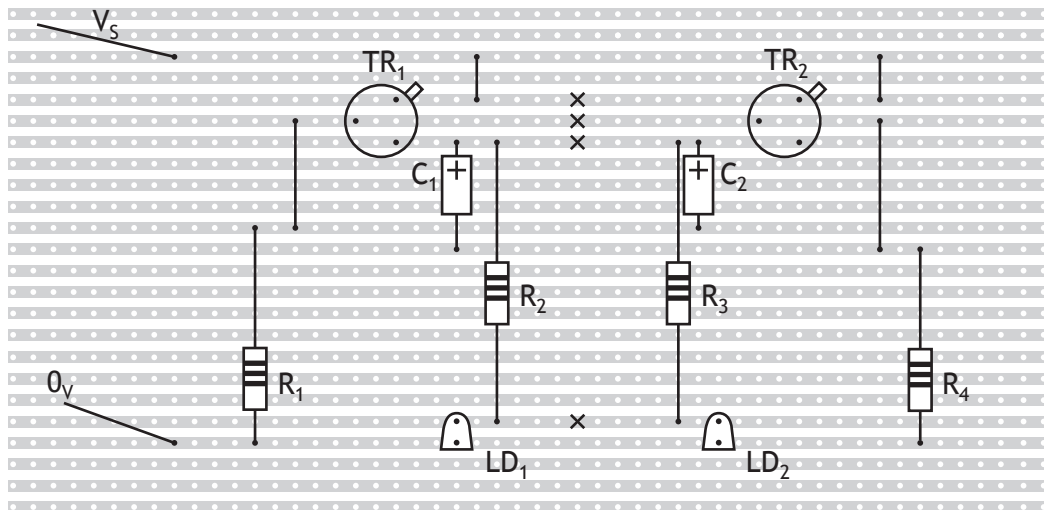
On your diagram, clearly indicate the input, process and output sections of your solution.

6



* X 8 6 0 7 5 0 1 1 7 *

10. The stripboard plan shows a **component (top) view** of a circuit with the following layout.



Component data

R_1 and R_4 – carbon film 2K2 0.25W

R_2 and R_3 – carbon film 390R 0.25W

C_1 – 100 μ F **electrolytic** capacitor

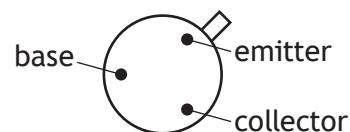
C_2 – 100 μ F **electrolytic** capacitor

LD_1 and LD_2 5mm standard LED

TR_1 and TR_2 – BC179 **pn**p bipolar transistors

X shows where a track has been cut

TR_1 and TR_2 pin connections



Draw a circuit diagram for this circuit.

Each component must be labelled.

6



* X 8 6 0 7 5 0 1 1 8 *

Question		Expected response	Max mark	Additional guidance
9.		1 mark for selecting both inputs 1 mark for selecting correct gates 1 mark for selecting both outputs 1 mark for correctly linking inputs to gates 1 mark for linking AND gate to outputs 1 mark correctly identifying the three sections of the system diagram	6	All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections).
10.		Both power rails labelled with nodes (1 mark) Transistor positions and type (npn used) (1 mark) Base to 0v via 2K2 resistors (1 mark) Collector to 0v via 390R resistor with LED correct orientation (1 mark) Both capacitor orientation and connections (1 mark) All components labelled (either values or identifiers) (1 mark)	6	If npn used and other connections are in comparable places then treat as repeated error and deduct 1 mark for transistor position/type Accept a mixture of identifiers and values.

[END OF MARKING INSTRUCTIONS]

Total marks — 60
Attempt ALL questions

1. The table below gives information about some circuit components.
Some of the boxes have been left blank.

(a) Complete the table for the missing entries.

3

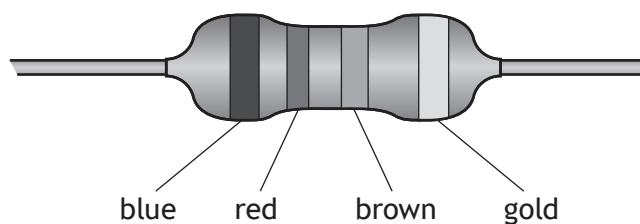
Component name	Symbol	Function
Bipolar transistor		electronic switch
Cell		
		measures voltage across a circuit
Lamp		converts electrical energy into light energy



* X 8 6 0 7 5 0 2 0 2 *

1. (continued)

(b) A pupil selects a resistor with the colour coding as shown below.



Use the information in the data sheet to answer the questions below.

- (i) Determine the resistance of the resistor. 1
- (ii) State the percentage tolerance of this resistor. 1
- (iii) Determine the maximum and minimum resistance of this resistor in ohms. 2

[Turn over

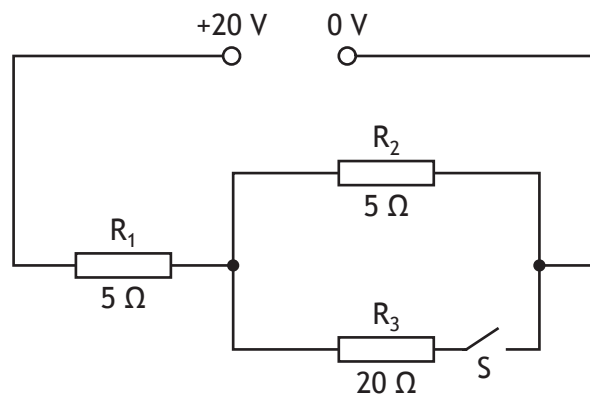


2. The table below shows information about the resistance of a thermistor at different temperatures.

Temperature (°C)	Resistance (kΩ)
20	50.0
40	30.0
60	
80	10.0

Complete the table above to predict the resistance of the thermistor at 60 °C.

3. A circuit diagram is shown below.



- (a) Switch S is **open**.

- (i) Calculate the total resistance of this circuit.

Space for working and answer

- (ii) Calculate the current in this circuit.

Space for working and answer

MARKS DO NOT WRITE IN THIS MARGIN

1

1

3



* X 8 6 0 7 5 0 2 0 4 *

Marking instruction for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Relay (1 mark) (1 mark)  Changes resistance with (changes in) temperature (1 mark)	3	Do not accept switch Accept alternatives:   Also accept -  Must have input & output lines. Accept heat Accept expressions involving TURD (Temperature goes Up, Resistance goes Down)
	(b)	(i)	7.5 kΩ OR 7K5 OR 7500 Ω	1	If given in ohms must have unit. Also accept scientific notation with unit Accept 7500R Don not accept 7.5KR
		(ii)	(±)10%	1	Accept answer without percentage If + or - is given on its own, do not award marks
		(iii)	Maximum = 8250 Ω (1 mark) Minimum = 6750 Ω (1 mark) 	2	Accept answer consistent with 1(b)(i). Accept in any order. Accept without Max & Min name, but if given must be correct response. No penalty for omitting units, as “in ohms” is stated in the question.
2.			Any two from • Wear eye protection • Use stand • Use heat proof mat • Avoid breathing in fumes • Hold soldering iron correctly	2	Any acceptable alternate answer Do not accept “remove clothing” needs to justify why

3. (a) (continued)

- (iii) Calculate the power dissipated in resistor
- R_1
- .

Space for working and answer

3

(b) Switch S is now **closed**.

- (i) Calculate the effective resistance of
- R_2
- and
- R_3
- in parallel.

Space for working and answer

3

- (ii) Calculate the overall circuit resistance.

Space for working and answer

1

[Turn over



* X 8 6 0 7 5 0 2 0 5 *

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \quad (1 \text{ mark})$ $\frac{1}{R_T} = \frac{1}{40} + \frac{1}{10} \quad (1 \text{ mark})$ $R_T = 8\Omega \quad (1 \text{ mark})$	3	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ Is an acceptable alternative method If wrong equation used, eg $R_T = \frac{1}{R_1} + \frac{1}{R_2}$ then (0) marks Accept imprecise working towards a final answer $\frac{1}{R_T} = 8\Omega$ ↑ accept Accept = 8 R Accept any subscript on component resistors
		(ii)	$(R_T = 8 + 4)$ $R_T = 12 \Omega$	1	Accept answer consistent with 3(a)(i) Accept 12R Must have correct unit
	(b)		$V = IR \quad (1 \text{ mark})$ $6 = I \times 12 \quad (1 \text{ mark})$ $I = 0.5 \text{ A} \quad (1 \text{ mark})$	3	Accept answer consistent with 3(a)(ii) 1 mark for both relationships 1 mark for substitutions 1 mark for final answer including unit Accept: $V_{R_3} = \left(\frac{R_3}{R_p + R_3} \right) V_s$ $V_{R_3} = \left(\frac{4}{8 + 4} \right) 6$ $= 2 \text{ V}$ And $V_{R_3} = I \times R_3 \quad 2 = I \times 4 \quad I = 2/4 = 0.5 \text{ A}$

Question			Expected response	Max mark	Additional guidance
3.	(c)		$P = I^2 R$ (1 mark) $P = 0.5^2 \times 4$ (1 mark) $P = 1W$ (1 mark)	3	Accept answer consistent with 3(b) with correct substitutions 1 mark for both relationships 1 mark for substitutions 1 mark for final answer including unit Accept $V = I R_3$ $0.5 \times 4 = 2 V$ $P = \frac{V^2}{R}$ $P = IV$ $P = \frac{2^2}{4}$ $P = 0.5 \times 2$ $P = 1W$ $P = 1W$

4. There are many different types of logic gate used in electronics.

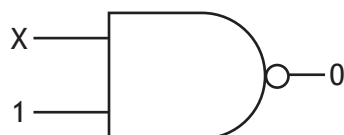
(a) Complete the truth table for an **NOR** gate.

1

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 20*.)

(b) A logic gate is shown below, with given logic states.



Determine the logic state of X that would produce the output shown.

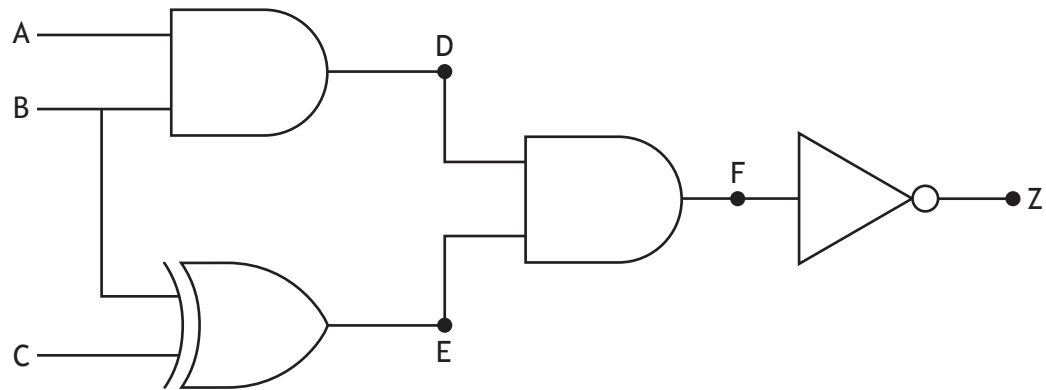
1



4. (continued)

(c) Complete the truth table for the logic circuit shown below.

4



A	B	C	D	E	F	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(An additional truth table, if required, can be found on *page 20*.)

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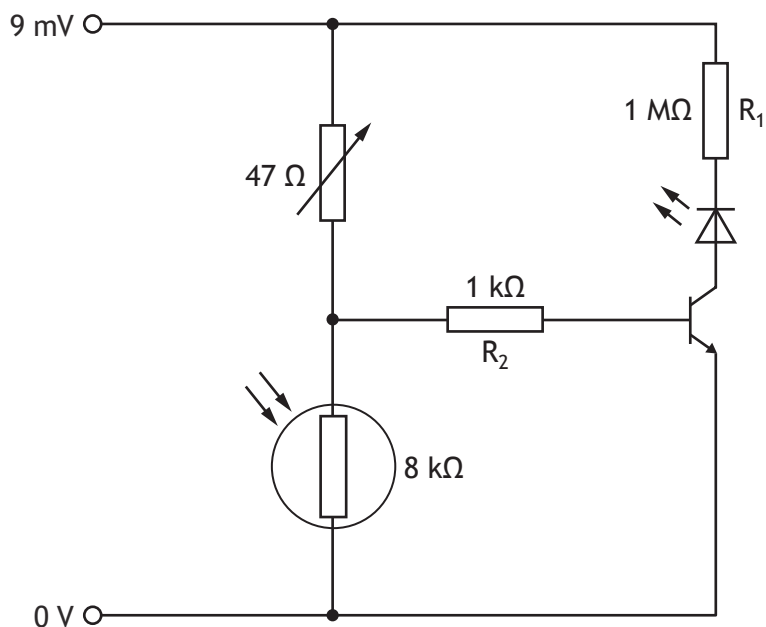
Question			Expected response	Max mark	Additional guidance																																																						
4.	(a)		<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Output	0	0	0	0	1	0	1	0	0	1	1	1	1																																								
A	B	Output																																																									
0	0	0																																																									
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1	1	1																																																									
	(b)		NOR (gate)	1																																																							
	(c)		<table><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table>	A	B	C	D	E	Z	0	0	0	1	1	1	0	0	1	1	0	1	0	1	0	1	1	1	0	1	1	1	0	1	1	0	0	1	1	1	1	0	1	1	0	1	1	1	0	0	1	1	1	1	1	0	0	0	3	1 mark per column Apply follow through between columns
A	B	C	D	E	Z																																																						
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1	1	1	0	0	0																																																						

5. A student produces a simulation of a circuit that will turn on an LED when it gets too cold in a greenhouse.

However, the simulation does not work as specified.

Identify **four** errors in the student's simulation shown below.

4



Error 1:

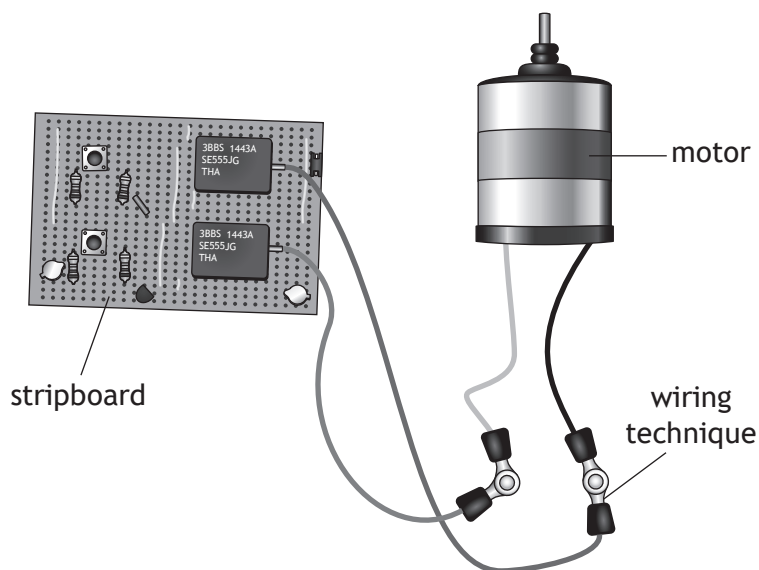
Error 2:

Error 3:

Error 4:



6. The image below shows the wiring connections between a stripboard and a motor for a circuit.



The wiring technique used in the photo is crimping.

- (a) State another wiring technique that could have been used to secure the wires together.
- (b) State one method that could be used to identify the correct wiring between the stripboard and motor.

1

1

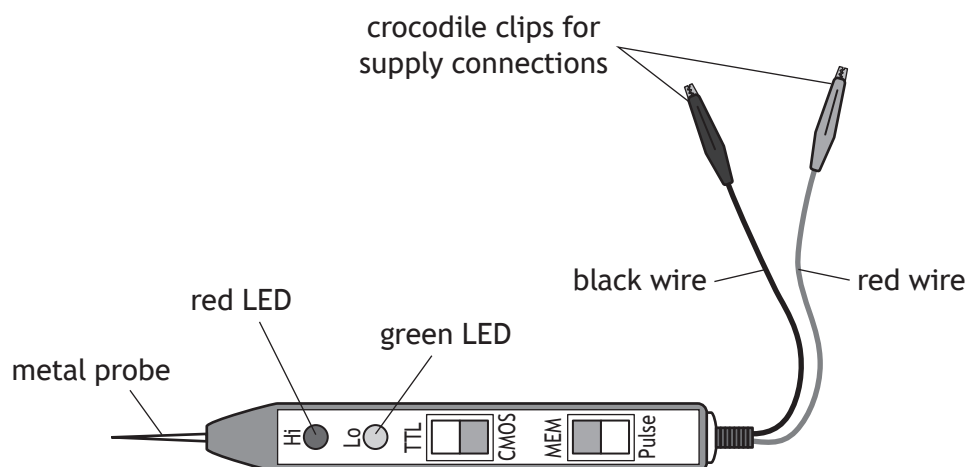
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Question			Expected response	Max mark	Additional guidance
5.	(a)		Any two from • To establish if the circuit works • To improve the circuit design • Easily edited to try different combinations • Allows testing of sub-systems	2	Any other suitable response Do not accept: allows you to cost the circuit
	(b)		Orientation of (electrolytic) capacitor or (electrolytic) capacitor wrong way round (1 mark) Diode used, not LED (1 mark) (Protective resistor) R3 (for LED) has too high a resistance/value (1 mark)	3	Errors can be listed in any order Accept not using a LED rather than a diode Do not accept resistor too big/large Do not accept pin 5 is not connected/floating
6.			Any three from • Check that there is no short circuit across the circuit supply terminals • Power supply correct way round (0V lower rail) • Transistors in correct position or location • Transistor orientations correct • Resistor values • Capacitor values correct • Resistors in correct location • Capacitors in correct location • Check wiring is secure • Check for solder bridges	3	Any other suitable response

7. A logic probe, shown below, is used to test the inputs and outputs of a 74 series logic chip.

The logic probe is set to TTL and pulse.



- (a) (i) State where the red and black wires should be connected.

1

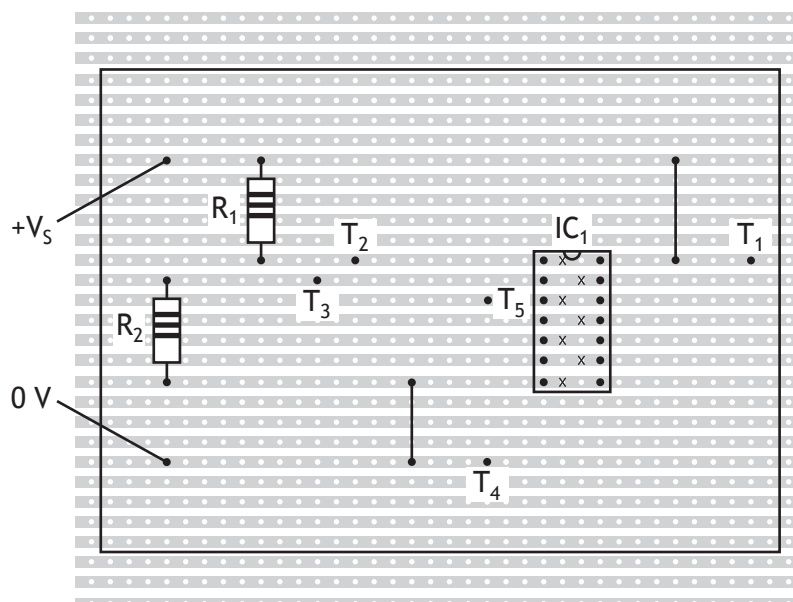
- (ii) Describe how a logic 1 would be detected.

2



7. (continued)

- (b) A student sets up a circuit on stripboard as shown below. IC₁ is a 74 series logic chip.



Test points T₁ to T₅ are marked on the stripboard.

- (i) Identify two test points that are used to measure the voltage across R₂.

1

- (ii) The logic probe is placed at pin 7 of the logic chip, IC₁.
Explain why the LED on the logic probe flashes green.

1

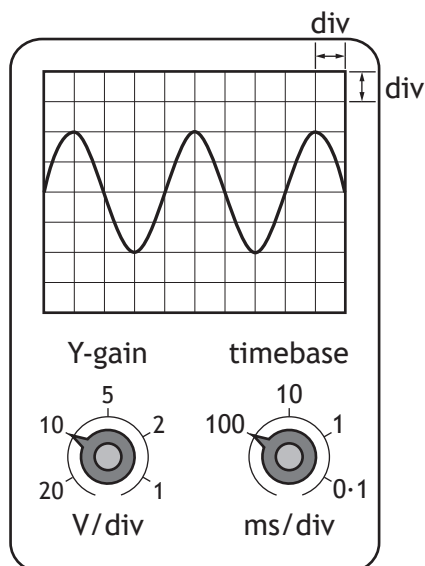
[Turn over]



8. The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are shown below.



- (a) Determine the peak voltage of this signal.

1

- (b) Calculate the frequency of the signal.

3

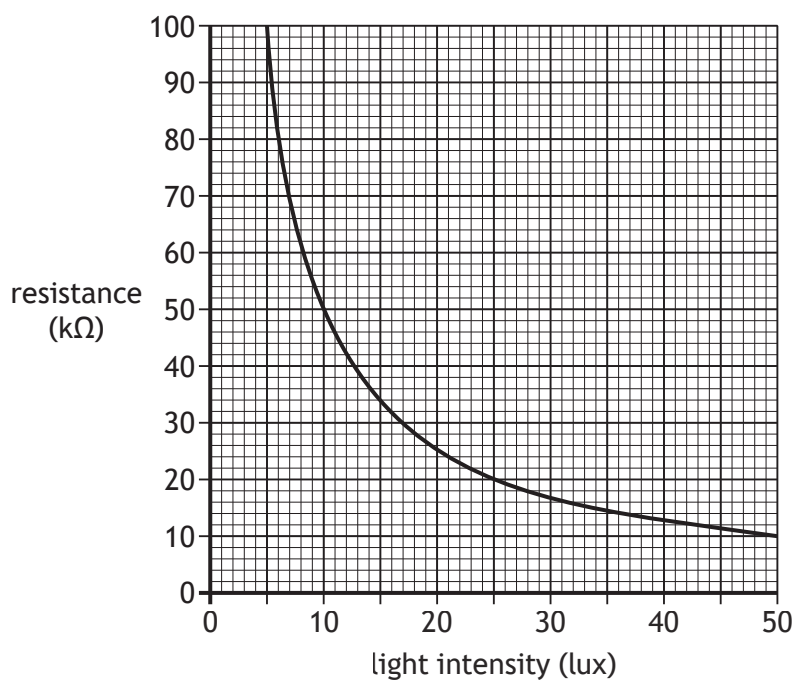
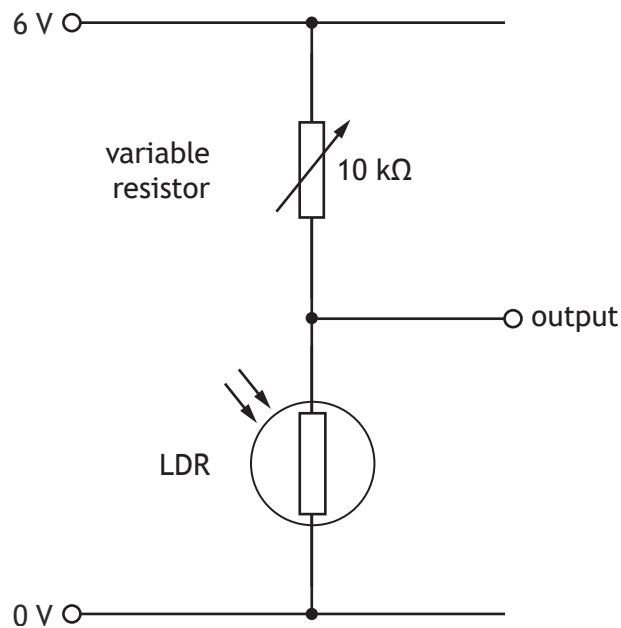
Space for working and answer



Question			Expected response	Max mark	Additional guidance
7.	(a)		Connect probe 1 to the VΩmA or middle socket of multimeter (1 mark)	2	Probe must be identified as probe 1 Accept Red instead of Probe 1 Accept range/scale as interchangeable
			Select 2000 μ range (1 mark)		
	(b)			1	Also accept A and symbol but do not accept V and symbol.
	(c)		20K	1	Accept 20,000 no unit required
8.	(a)		Analogue (signal)	1	
	(b)		4 V	1	Must include unit.
	(c)		$T = (4 \times 1 \text{ ms}) = 4 \text{ ms}$ or $4 \times 10^{-3} \text{ s}$ or 0.004 s (1 mark) $f = \frac{1}{T}$ (1 mark) $f = 250 \text{ Hz}$ (1 mark)	3	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula If period incorrect then award a maximum of 1 mark for the formula If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0.25 kHz , in which case 3 marks may be awarded Accept $f=N/t$

9. An LDR is used as a light sensor in a circuit to monitor and control the automatic mode setting of a car's headlights. Part of the circuit is shown below.

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WRITE IN
THIS
MARGIN



- (a) (i) The graph above shows how the resistance of the LDR varies with the light intensity, measured in lux.

Use the graph to determine the resistance of the LDR when the light intensity is 25 lux.

1

9. (a) (continued)

MARKS

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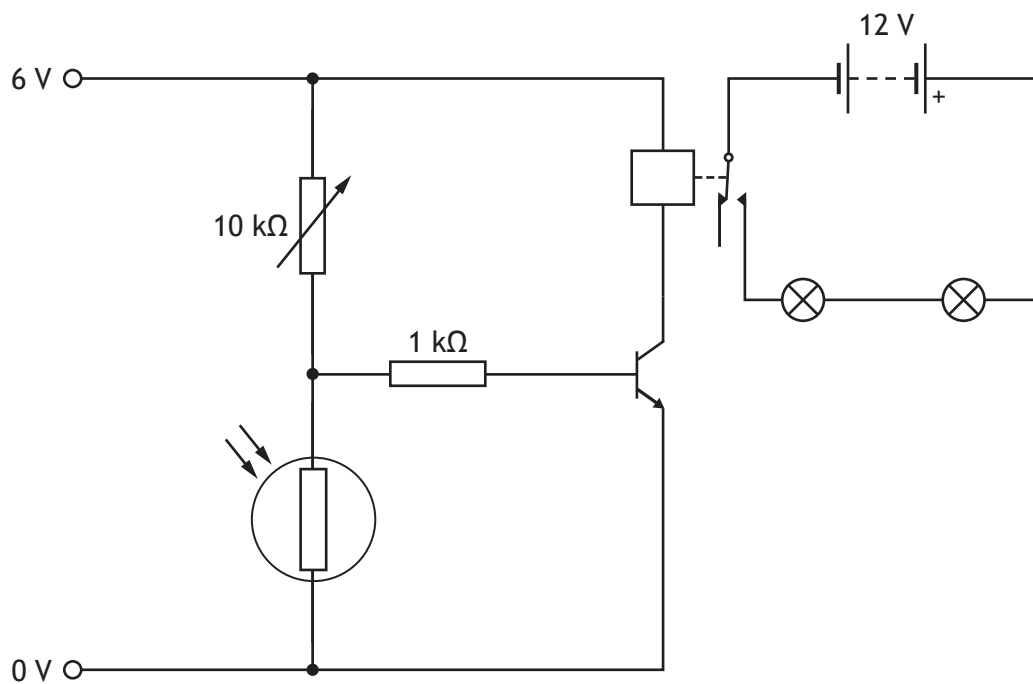
- (ii) The variable resistor is now set at a resistance of $10\text{ k}\Omega$.

Calculate the voltage across the LDR when the light intensity is kept at 25 lux.

3

Space for working and answer

- (b) The potential divider is now connected to a switching circuit via the car battery to operate the car lights, as shown below.



- (i) Explain how the circuit operates in order to switch on the car lights when the light intensity falls below a certain value.

3



* X 8 6 0 7 5 0 2 1 5 *

9. (b) (continued)

- (ii) The operation of the circuit can be improved by the addition of one component connected in parallel with the relay.

Name this component.

1



* X 8 6 0 7 5 0 2 1 6 *

Question			Expected response	Max mark	Additional guidance
9.	(a)		$V_x = \frac{R_2}{R_1 + R_2} \times V_s \quad (1 \text{ mark})$ $V_x = \frac{4000}{(8000 + 4000)} \times 6 \quad (1 \text{ mark})$ $V_x = 2V \quad (1 \text{ mark})$	3	Accept $(4/12) \times 6$ 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit Or by voltage ratio method $V_x / V_s = R_2 / R_{\text{total}}$ $V_x / 6 = 4000 / 12000$ $V_x = 2V$ If correct answer with unit and no working award 3 marks
	(b)		$6 \text{ k}\Omega$ OR 6000Ω OR $6K$	1	Must have unit or be in R notation Accept answer consistent with V_x calculated in 9(a)
	(c)		As light level reduces LDR resistance increases (1 mark) As LDR resistance increases the voltage across LDR increases (1 mark) When voltage across the LDR = V_x then LM741/comparator/IC/op amp output switches on, powering the LED (1 mark)	3	Accept $V_{\text{ref}} = V_x = \text{trigger voltage} = \text{switch on voltage}$ Accept marks independently Accept LED lights up Accept alternatives to = Do not accept arrow symbols as alternatives to increase and decrease

10. A gas pipeline mechanism is required to detect gas leaks.

When a leak is detected, the gas should be switched off by a motor.

The system must include a gas sensor to monitor for any leaks.

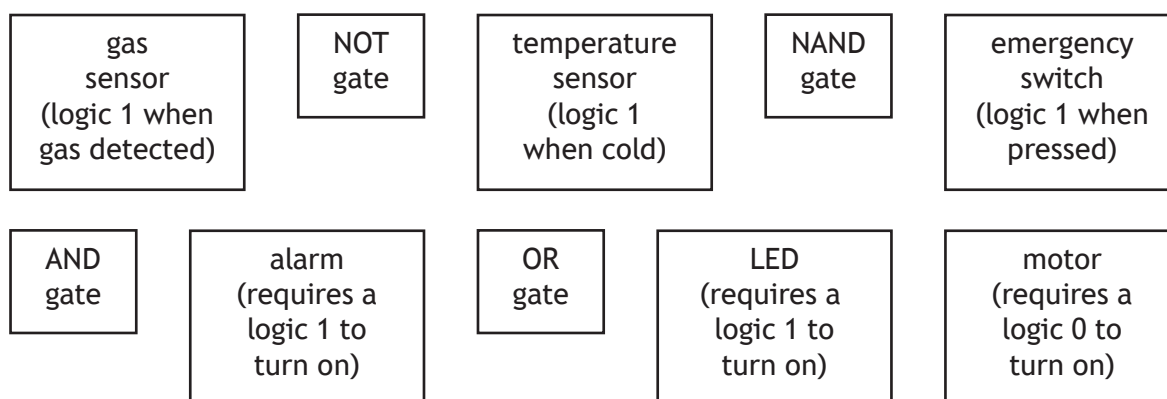
The gas can be switched off when either an engineer presses an emergency switch or when the gas sensor detects gas.

If the gas is switched off, an audible alert should also be sounded.

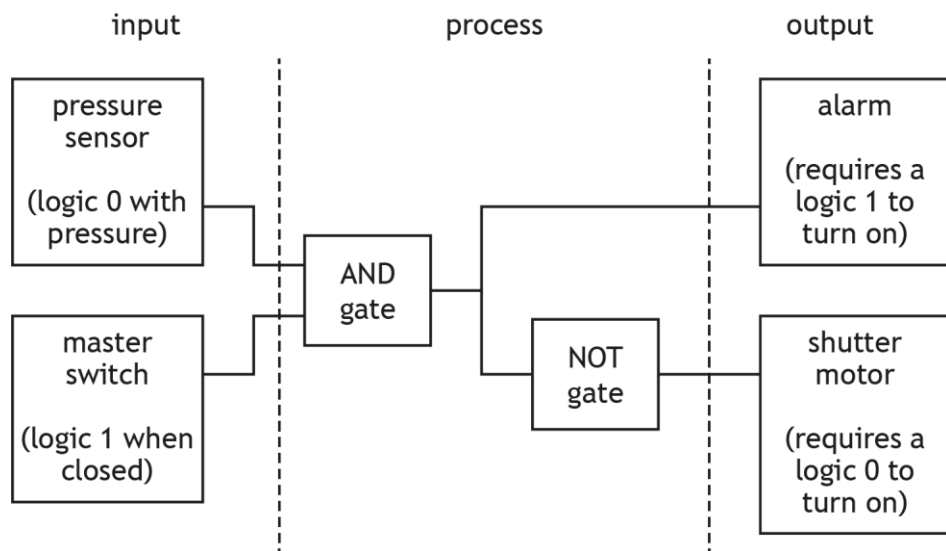
Selecting from the elements given below, draw a block diagram of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution.

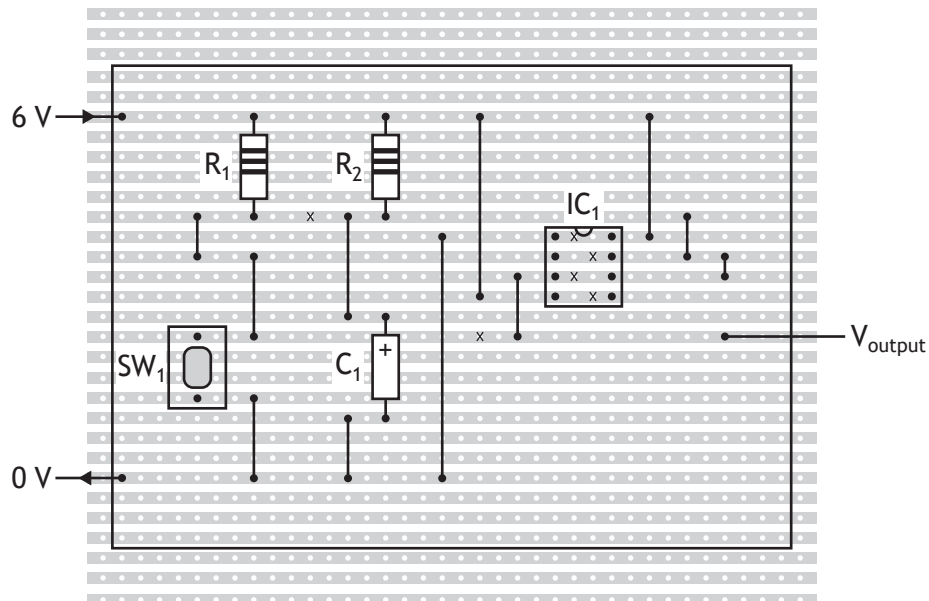
6



Question		Expected response	Max mark	Additional guidance
10.		1 mark for selecting both input devices 1 mark for selecting correct gates 1 mark for selecting both outputs 1 mark for correctly linking the inputs and gates 1 mark for linking both AND gate and NOT gate to outputs 1 mark for correctly identifying the three sections of the system	6	All selected elements must be shown as part of a system/block diagram (arrows are NOT required on the connections) Accept switch, alarm, motor and pressure sensor as block descriptors Accept solid separation lines Do not accept a list/table of elements



11. The stripboard plan below shows a **component (top) view** of a circuit with the following layout.



Component data

IC_1 — NE555

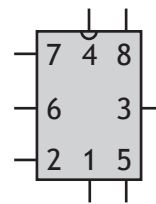
R_1 and R_2 — carbon film 10 K Ω 0.25 W

C_1 — 1000 μ F 16 V **electrolytic** capacitor

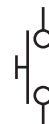
SW_1 — push-to-make switch

X shows where a track has been cut.

Circuit symbol for IC_1



Circuit symbol for SW_1

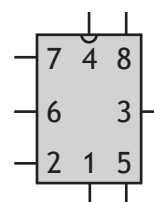


11. (continued)

For the stripboard layout diagram, complete the circuit diagram below.

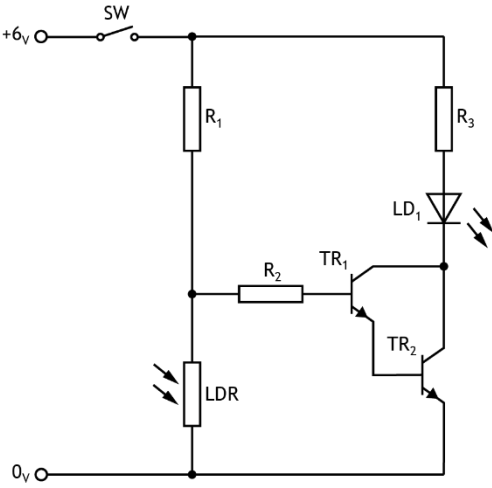
Each component must be labelled.

(An additional circuit diagram template, if required, can be found on *page 21*.)



[END OF QUESTION PAPER]



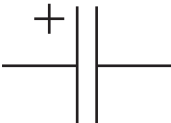
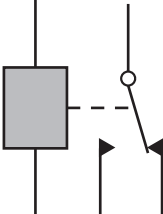
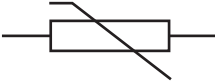
Question		Expected response	Max mark	Additional guidance
11.		Both power rails labelled with nodes (1 mark) Switch in correct position (1 mark) R_1, LDR voltage divider to R_2 (1 mark) Transistor positions/connections in relation to each other and 0v rail. (1 mark) Collectors to + 6V via 270R resistor and LED in series with correct orientation. (1 mark) Component labelling (either values or identifiers) (1 mark)	6	 Any errors in circuit symbols max 5 marks, only if candidate has achieved maximum marks for the question Accept LED₁ as the label Accept SPST as a descriptor for SW

[END OF MARKING INSTRUCTIONS]

Total marks — 60
Attempt ALL questions

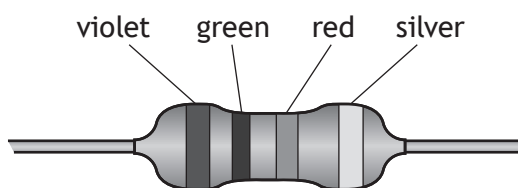
1. The table below gives information about some circuit components.
Some of the boxes have been left blank.
- (a) Complete the table for the missing entries.

3

Component name	Symbol	Function
Electrolytic capacitor		stores charge
		electromagnetic switch
Diode		allows current to flow in one direction only
Thermistor		

1. (continued)

(b) A technician selects a resistor with the colour coding as shown below.



Use the information in the data sheet to answer the questions below.

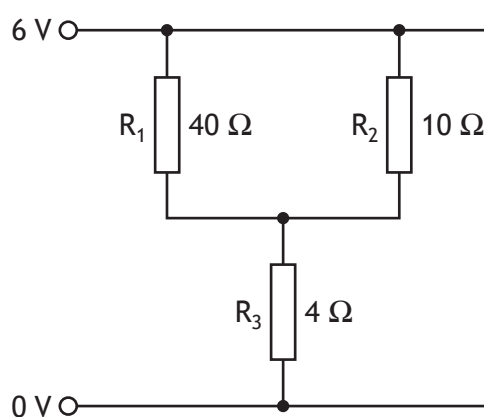
- (i) Determine the resistance of this resistor. 1
- (ii) State the percentage tolerance of this resistor. 1
- (iii) Determine the maximum and minimum resistance of this resistor in ohms. 2



2. State **two** safety precautions that must be taken when using a soldering iron.

2

3. A circuit diagram is shown below.



- (a) (i) Calculate the effective resistance of R_1 and R_2 in parallel.
Space for working and answer

3

- (ii) Calculate the total resistance of this circuit.
Space for working and answer

1



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3. (continued)

(b) Calculate the current in resistor R_3 .

3

Space for working and answer(c) Calculate the power dissipated in resistor R_3 .

3

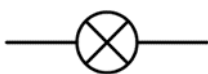
Space for working and answer

[Turn over



* X 8 6 0 7 5 0 2 0 5 *

Marking instruction for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Supplies electrical (potential) energy/charge (1 mark)	3	Do not accept: - Supplies power Power supply Stores power Also accept: - Converts chemical energy to electrical
			voltmeter (1 mark)		Do not accept old bulb symbol
			 (1 mark)		
	(b)	(i)	620Ω (1 mark) OR 620R OR 0.62kΩ	1	If given in ohms must have unit, otherwise 0 marks. Also accept scientific notation with unit. $6.2 \times 10^2 \Omega$
		(ii)	(±)5(%)	1	Accept answer without tolerance and/or percentage. Do not accept + or - on its own.
		(iii)	(Maximum) = 651Ω OR 651R (1 mark) (Minimum) = 589Ω OR 589R (1 mark)	2	Accept answer consistent with 1(b) (i) & (ii). Accept in any order. Accept value only. If maximum and minimum is incorrectly labelled award 0 marks.
2.			Any value greater than 10.0kΩ and less than 30.0kΩ (1 mark)	1	Accept a value only. Award correct mark if answer is out with table but must have correct unit.

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	$(R_T = 5 + 5)$ (1 mark) $R_T = 10 \Omega$	1	Accept 10R Must have correct unit.
		(ii)	$V = IR$ (1 mark) $20 = I \times 10$ (1 mark) $I = 2 \text{ A}$ (1 mark)	3	Accept answer consistent with 3(a)(i) 1 mark for both relationships. 1 mark for substitutions. 1 mark for final answer including unit. Also accept alternative methods. Do not accept the term “Amps” as units.
		(iii)	$P = I^2 R$ (1 mark) $P = 2^2 \times 5$ (1 mark) $P = 20W$ (1 mark)	3	Accept answer consistent with 3(a)(ii) 1 mark for relationship. 1 mark for substitutions. 1 mark for final answer including unit. Also accept alternative formulas but with correct substitution.
	(b)	(i)	$\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3}$ (1 mark) $\frac{1}{R_T} = \frac{1}{5} + \frac{1}{20}$ (1 mark) $R_T = 4\Omega$ (1 mark)	3	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ Is an acceptable alternative method. If wrong equation used, eg, $R_T = \frac{1}{R_1} + \frac{1}{R_2}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = 4\Omega$  accept Accept = 4 R Accept any subscript on component resistors.

Question			Expected response	Max mark	Additional guidance
3.	(b)	(ii)	$(R_T = 5 + 4)$ $R_T = 9 \Omega$	1	Accept answer consistent with 3(b)(i) Accept 9R Must have correct unit.

4. There are many different types of logic gate used in electronics.

(a) Complete the truth table for an AND gate.

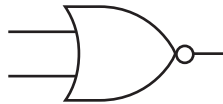
1

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 16*.)

(b) Name the logic gate shown below.

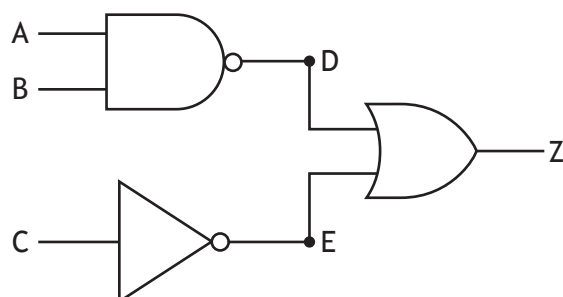
1



4. (continued)

(c) Complete the truth table for the logic circuit shown below.

3



A	B	C	D	E	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 16*.)

[Turn over



* X 8 6 0 7 5 0 2 0 7 *

Question			Expected response	Max mark	Additional guidance																																																															
4.	(a)		<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	1																																																	
A	B	Output																																																																		
0	0	1																																																																		
0	1	0																																																																		
1	0	0																																																																		
1	1	0																																																																		
	(b)		(Logic State) 1 (1 mark)	1	Accept 1, High, ON																																																															
	(c)		<table><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></table>	A	B	C	D	E	F	Z	0	0	0	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	1	0	1	1	0	0	0	1	1	0	0	0	0	0	1	1	0	1	0	1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0	1	4	1 mark per column. Apply follow through between columns.
A	B	C	D	E	F	Z																																																														
0	0	0	0	0	0	1																																																														
0	0	1	0	1	0	1																																																														
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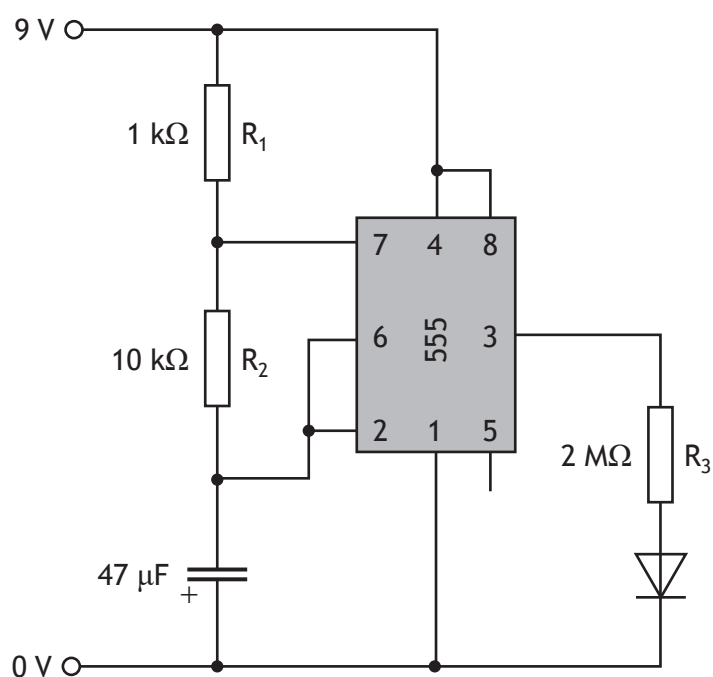
5. Simulation is used in circuit design.

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(a) State **two** reasons for simulating a circuit before it is constructed.

2

(b) A student simulated the flashing LED circuit shown below.



Identify **three** errors in the student's circuit.

3

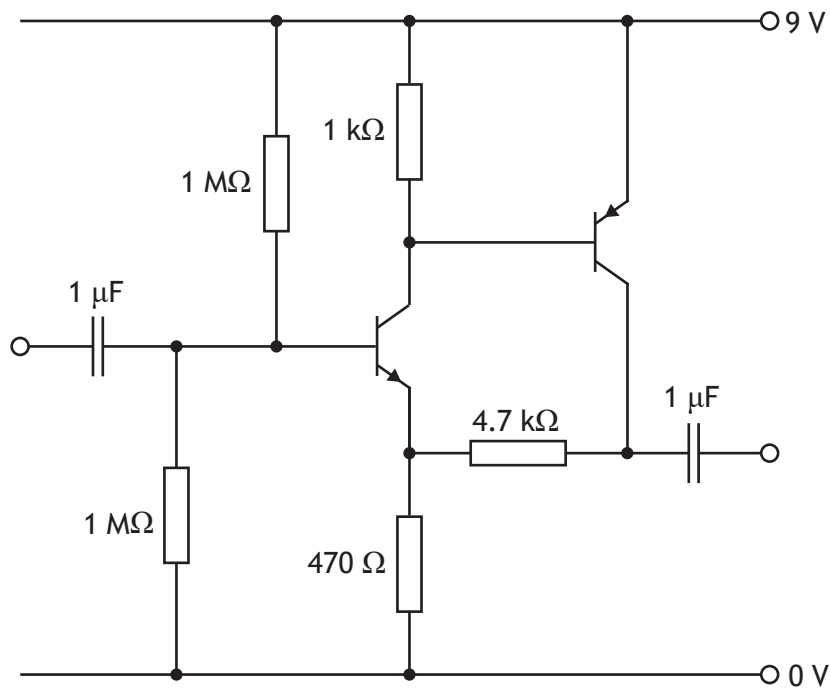
Error 1:

Error 2:

Error 3:



6. A technician constructed a circuit from the schematic diagram shown below.



Complete the following table by giving **three** pre-power up checks for this circuit.

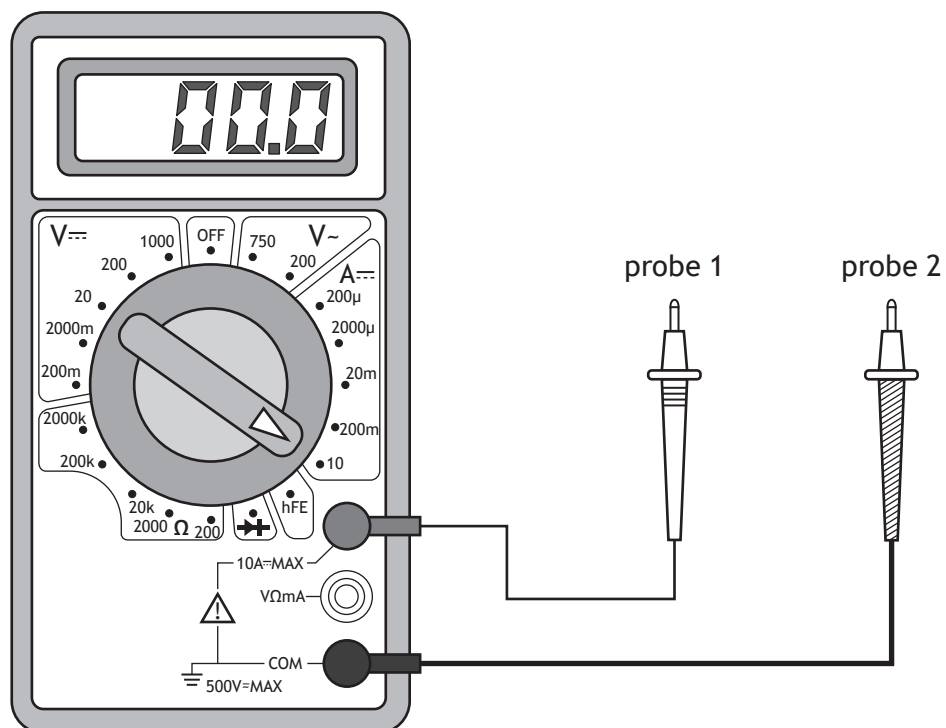
3

Pre-power up checklist
Correct power supply voltage

[Turn over

Question			Expected response	Max mark	Additional guidance
5.			Any four from • voltage supply value too small. • LED orientation/polarity wrong way round. • protective resistor (R_1) for LED has too high a resistance value. • incorrect component symbol used to measure temperature (Thermistor not LDR).	4	Errors can be listed in any order. Do not accept resistor too big/large.
6.	(a)		Connector blocks (1 mark) OR Terminal blocks	1	Do not accept the following: Cable ties, heat shrink, velcro straps, spiral wrap or connector on its own. Accept Solder and heat shrink. Accept screw terminal.
	(b)		Colour coding (1 mark) OR Wire/cable numbering	1	Any other suitable response.

7. A student wants to check that the current flowing through a component is 1.2 mA DC.



- (a) Identify **two** changes that must be made to the multimeter to get the most accurate measurement.

2

Change 1:

Change 2:

- (b) Draw the symbol for direct current that is shown on the meter.

1

- (c) State the most appropriate scale that should be used to check the resistance of a 15 000 Ω resistor.

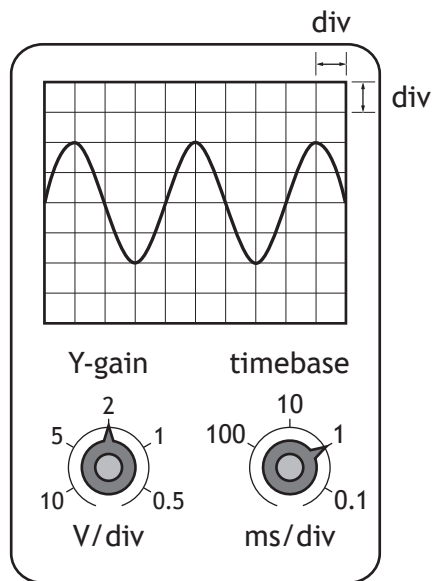
1



8. The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown below.



- (a) State whether this signal is analogue or digital.

1

- (b) Determine the peak voltage of this signal.

1

- (c) Calculate the frequency of this signal.

3

Space for working and answer

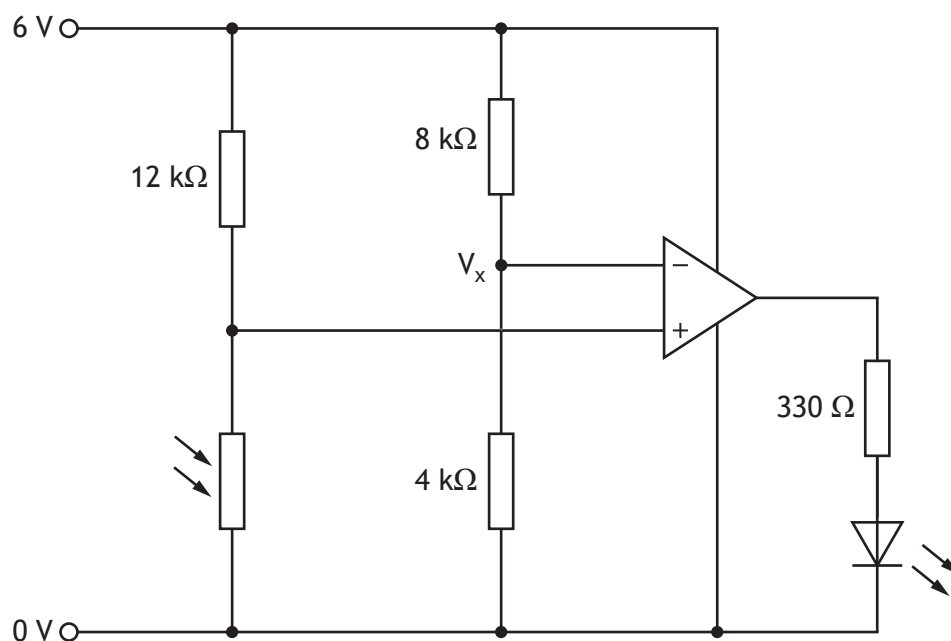


* X 8 6 0 7 5 0 2 1 1 *

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	Connect red wire of probe to +V _{supply} and black wire to 0v or equivalent. (1 mark)	1	Do not accept: connect the probe to the supply rails (on its own). Accept alternatives to V _{supply} , +V _s , +9V
		(ii)	Place the probe tip onto circuit at selected point. (1 mark) Red LED lights. (1 mark)	2	Independent marks.
	(b)	(i)	T3 and T4 (1 mark)	1	Answer must state test point numbers.
		(ii)	Pin 7 is connected to 0v or ground. (1 mark)	1	Accept: negative terminal (of the supply). Do not accept, “connected to the power supply” as it does not state high or low potential. Do not accept, “connected directly to the power supply”. Do not accept “PIN 7 is logic 0 or negative” on its own.
8.	(a)		20 V (1 mark)	1	Must include unit.
	(b)		$T = (4 \times 100 \text{ ms}) = 400 \text{ ms}$ or 0.4s (1 mark) $f = \frac{1}{T}$ (1 mark) $f = 2.5 \text{ Hz}$ (1 mark)	3	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0.0025 kHz, in which case 3 marks may be awarded. Do not accept the term “Secs” as unit for seconds.

9. A child's LED night light is controlled using a LM741 comparator circuit as shown below.

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- (a) Calculate the reference voltage V_x .

3

Space for working and answer

- (b) State the resistance of the light dependent resistor when the voltage across it is equal to the reference voltage V_x .

1

- (c) Describe how this circuit works.

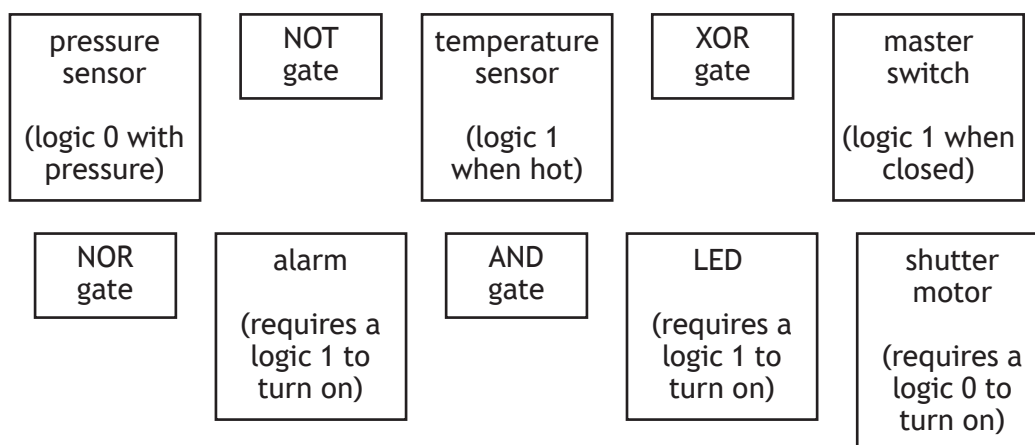
3



Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	20kΩ (1 mark) OR 20000Ω OR 20K	1	Must have unit or be in R notation. Allow a tolerance of + or - half a box.
		(ii)	$V_2 = \frac{R_2}{R_1 + R_2} \times V_s$ (1 mark) $V_2 = \frac{20}{(20+10)} \times 6$ (1 mark) $V_2 = 4V$ (1 mark)	3	Accept (20/30) × 6 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. $V_2/V_s = R_2/R_{total}$ $V_2/6 = 20/30$ $V_2 = 4V$ If correct answer with unit and no working award 3 marks. Accept follow through rule from Q9 a(i).
	(b)	(i)	LDR resistance increases and voltage across it increases. (1 mark) Transistor switches on at a certain voltage/0.7v (1 mark) Relay activates/turns on/closes (1 mark)	3	Independent marks Accept Transistor switches on/switch on voltage at 0.7v Answer must include the following three key points: - • LDR resistance increases • voltage increases to turn on Transistor at certain voltage • turns on RELAY (external circuit switches on) If transistor switching voltage is given as 2v award a maximum of 2 marks.
		(ii)	Diode (1 mark)	1	Do not accept the circuit symbol, the name must be given.

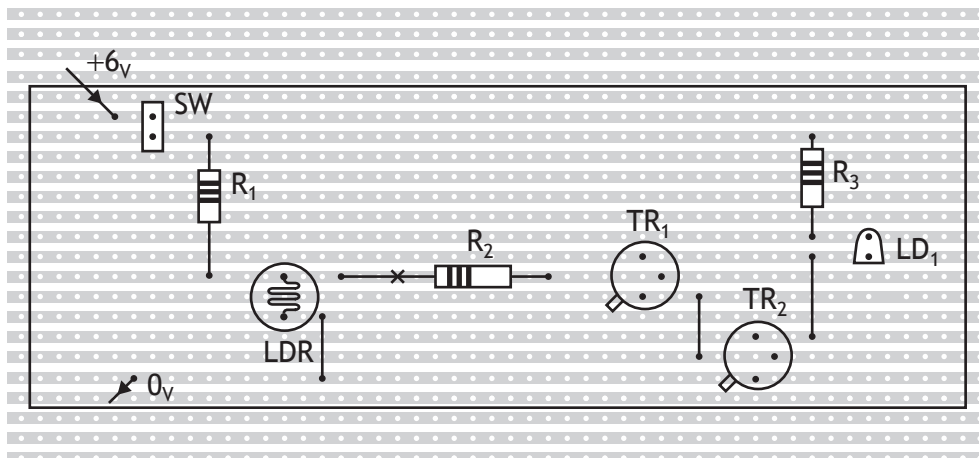
10. A security system for an art gallery is required to protect an expensive vase. The system must include a sensor that can detect if the vase has been lifted. When the vase is lifted, this will turn on an alarm and a motor which closes the shutter on the gallery door. There is also a master switch to turn the whole system on and off. Selecting from the elements given below, draw a **block diagram** of an electronic solution for this system. On your diagram, clearly indicate the input, process and output sections of your solution.

6



Question			Expected response	Max mark	Additional guidance
10.			1 mark for correctly identifying the three separate sections of the system. 1 mark for selecting both input devices. 1 mark for selecting correct gates. 1 mark for selecting both outputs. 1 mark for correctly linking the inputs to the gates. 1 mark for correctly linking the outputs to the gates.	6	All selected elements must be shown as part of a system/block diagram. (Arrows are NOT required on the connections). Accept switch, alarm, motor and gas sensor as block descriptors. Max 5 marks if not block diagram. Do not accept logic gate symbols.

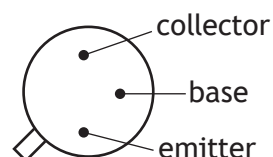
11. The stripboard plan below shows a **component (top) view** of a circuit with the following layout.



Component data

TR₁ and TR₂ — BFY51 npn transistors
 LD₁ — 5 mm standard led (red)
 LDR — ORP12 light dependent resistor
 R₁ — carbon film 10 K 0.25 w
 R₂ — carbon film 6K8 0.25 w
 R₃ — carbon film 270 R 0.25 w
 SW — SPST switch (on/off switch)

TR₁ and TR₂ pin connections

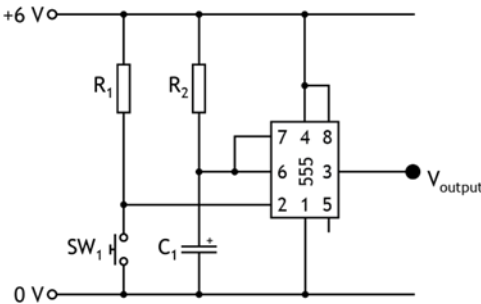


Draw a circuit diagram for this circuit.
 Each component must be labelled.

6



* X 8 6 0 7 5 0 2 1 4 *

Question			Expected response	Max mark	Additional guidance
11.			Both power rails labelled with nodes. (1 mark) Power rails to IC. (4,8 to +6V and 1 to 0V) (1 mark) SW₁ and R₁ in series between power rails, correctly positioned. (1 mark) R₂ and C₁ in series between power rails with C₁ correctly orientated. (1 mark) Pins 6, 7 and 2 connected, 5 not connected and correct connection of V_{output} to pin 3 of 555 timer. Node must be included. (1 mark) Each component must be identifiable and correctly used circuit symbols. (accept either values or identifiers). (1 mark)	6	 Accept IC1, 555 or NE555 for chip identifier.

[END OF MARKING INSTRUCTIONS]